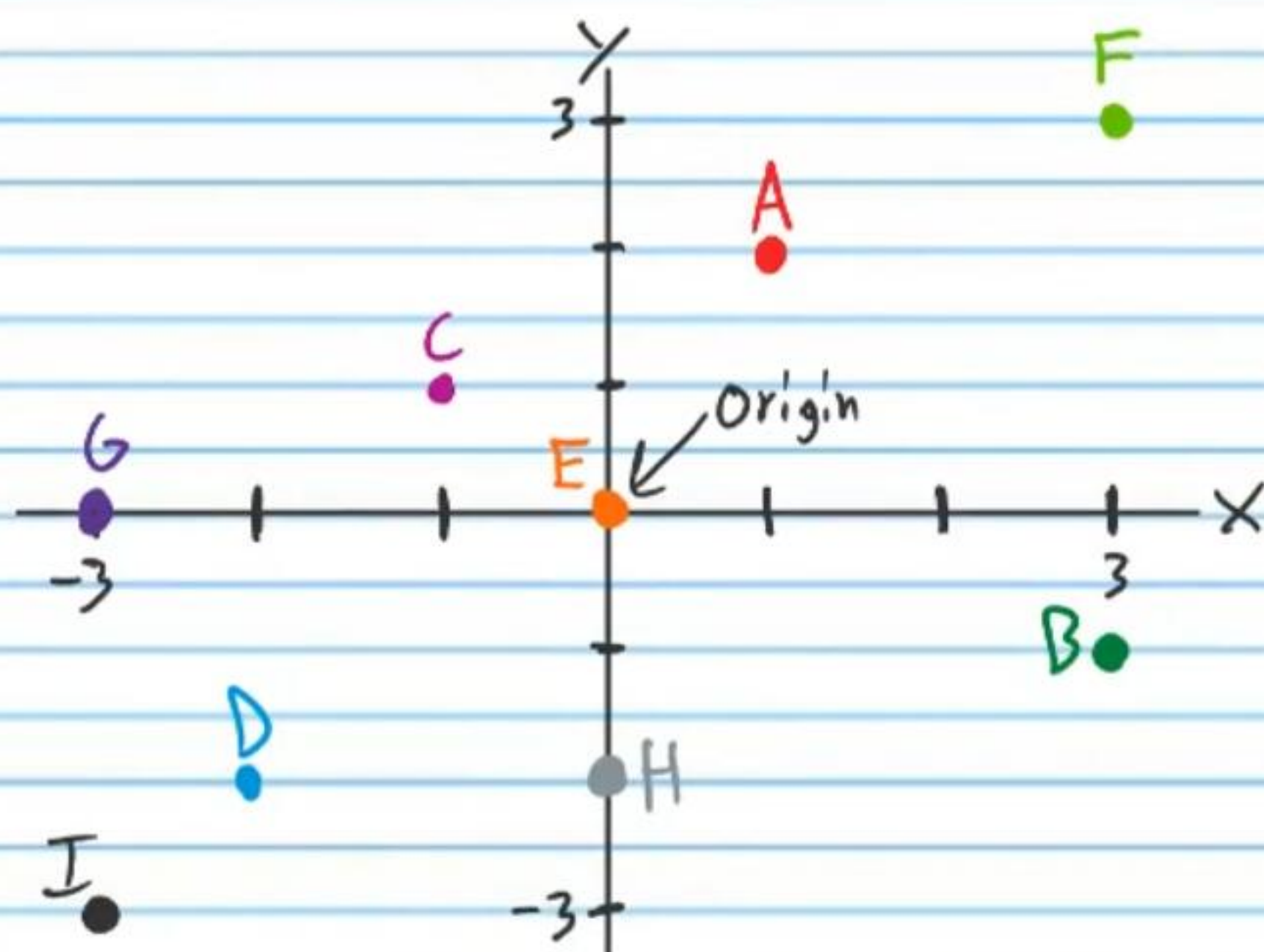


Graphing Linear Equations and Inequalities

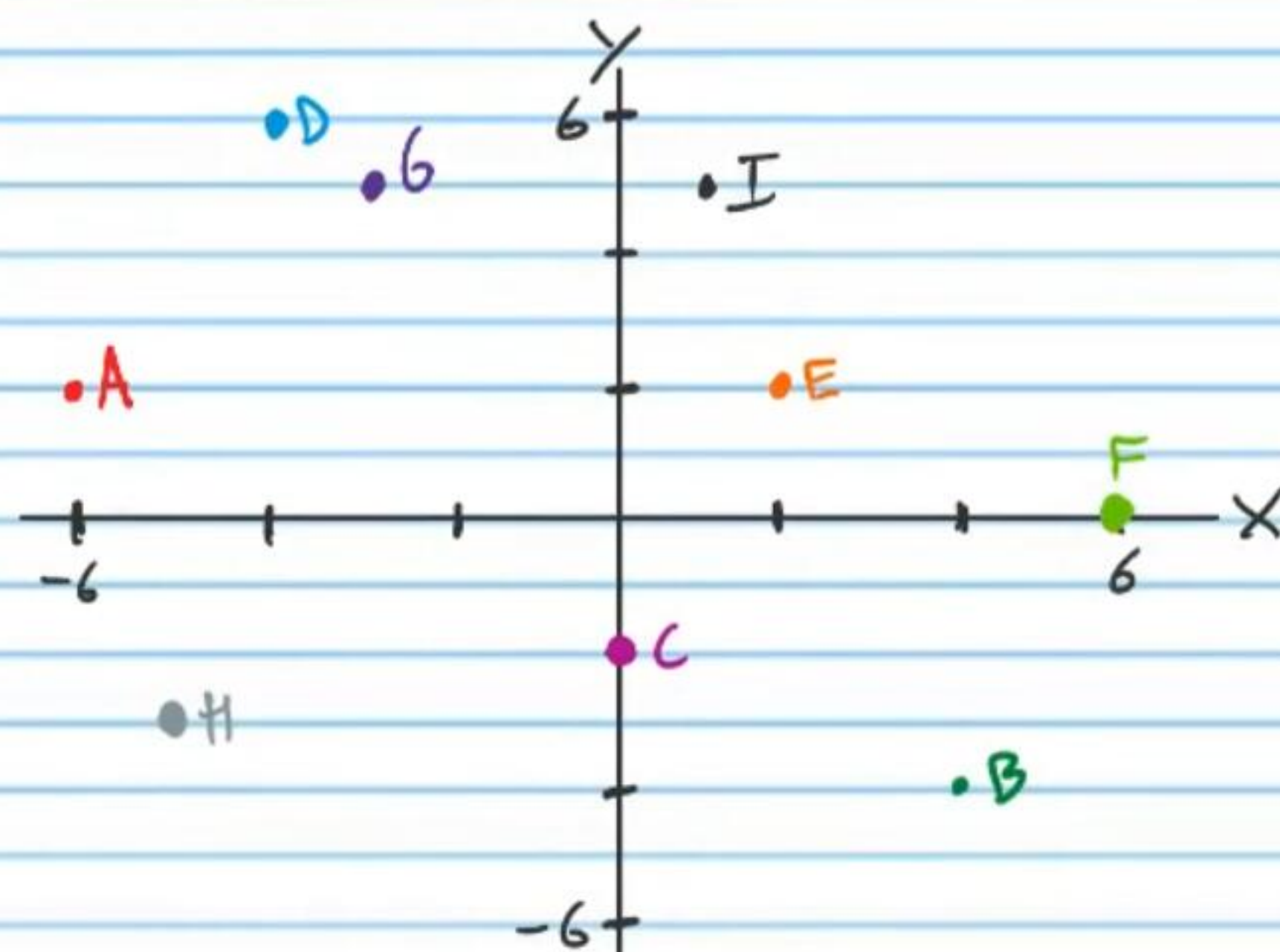
Name
Date
Course

Plotting Points

A(1,2)
B(3,-1)
C(-1,1)
D(-2,-2)
E(0,0)
F(3,3)
G(-3,0)
H(0,-2)
I(-3,-3)



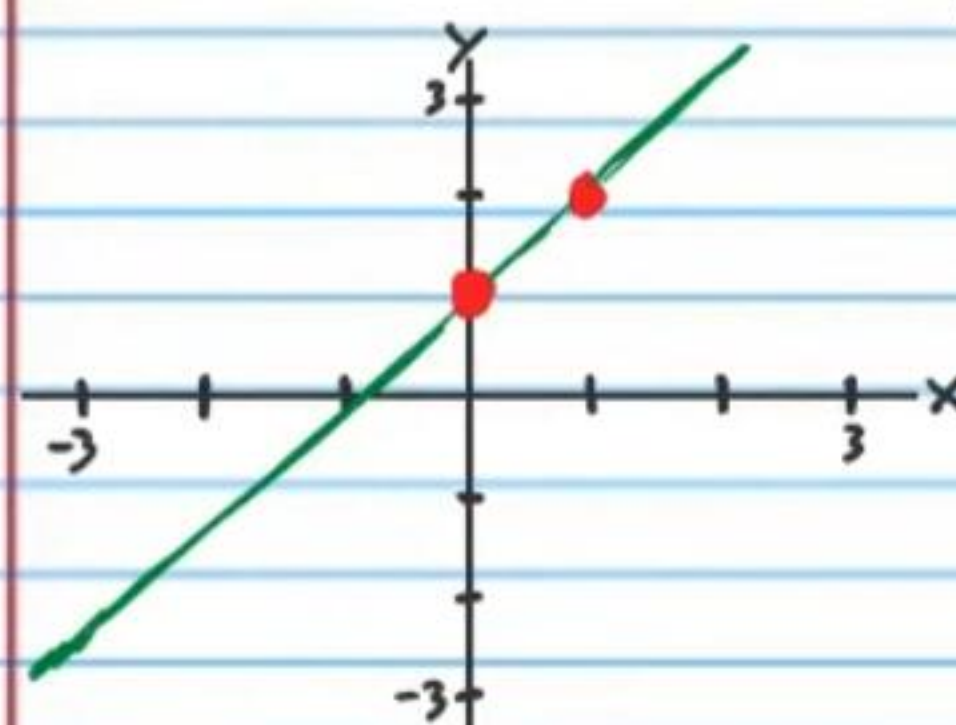
A(-6,2)
B(4,-4)
C(0,-2)
D(-4,6)
E(2,2)
F(6,0)
G(-3,5)
H(-5,-3)
I(1,5)



Graphing Linear Equations.

① $y = x + 1$

x	y
0	1
1	2



1st Point

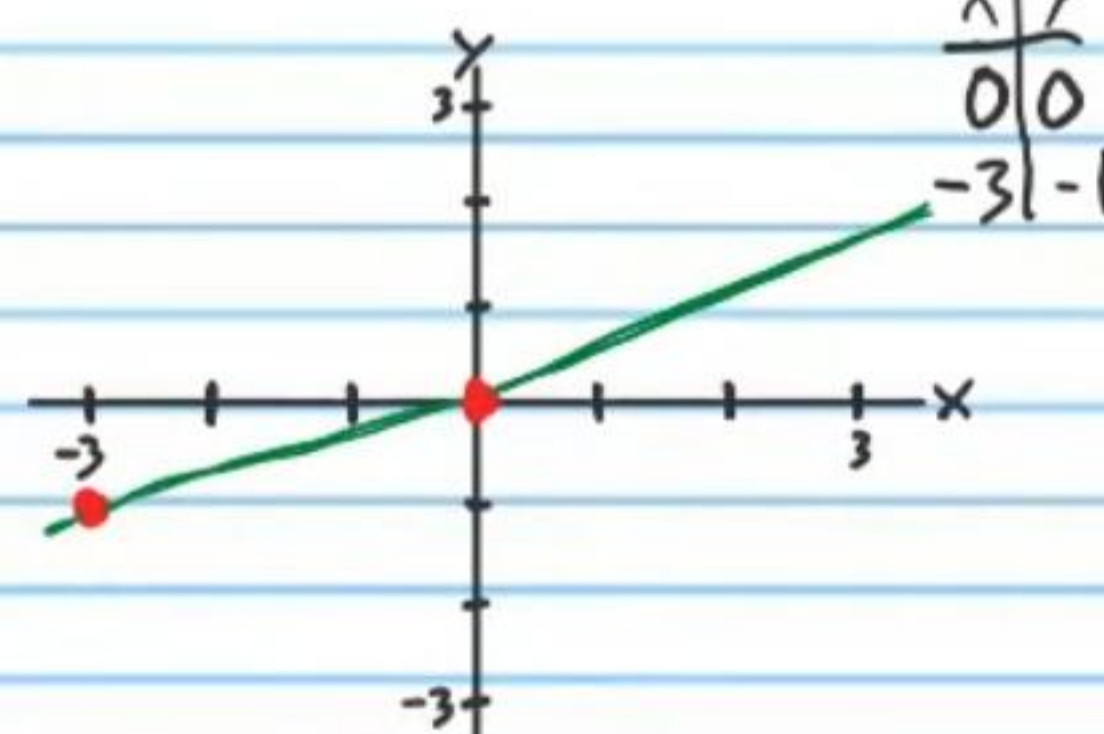
$x = 0$
↓
 $y = 0 + 1$
 $y = 1$
(0,1)

2nd Point

$x = 1$
↓
 $y = 1 + 1$
 $y = 2$
(1,2)

② $x = 3y$

x	y
0	0
-3	-1



1st Point

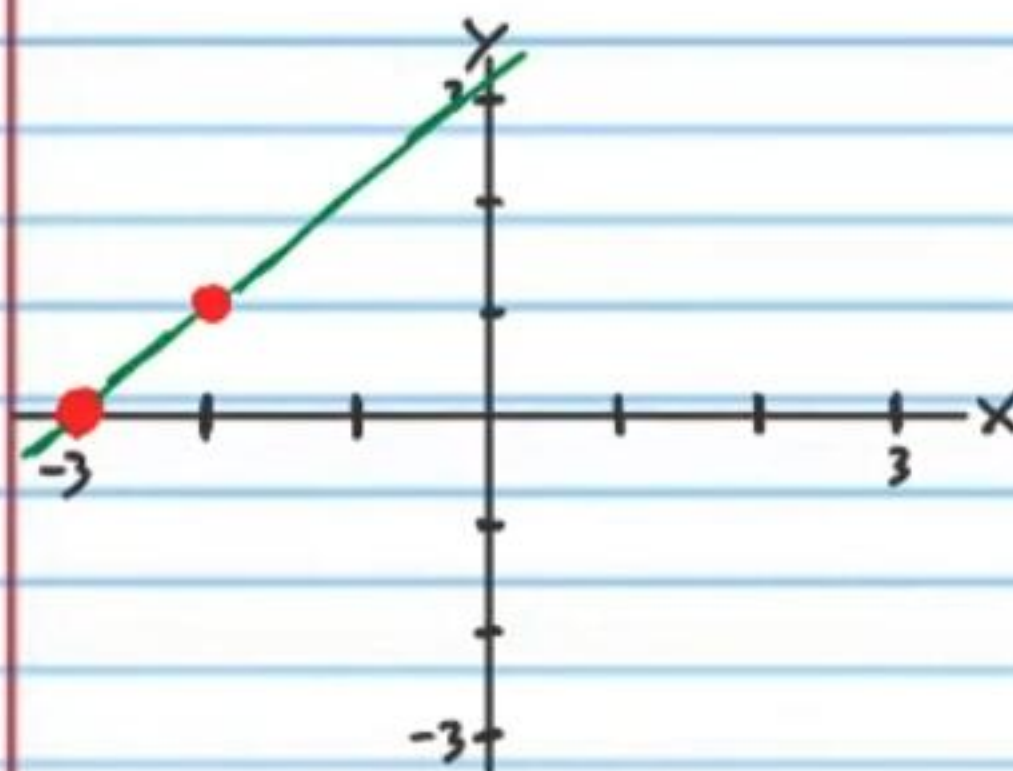
$y = 0$
↓
 $x = 3(0)$
 $x = 0$
(0,0)

2nd Point

$y = -1$
↓
 $x = 3(-1)$
 $x = -3$
(-3,-1)

③ $x = y - 3$

x	y
-3	0
-2	1



1st Point

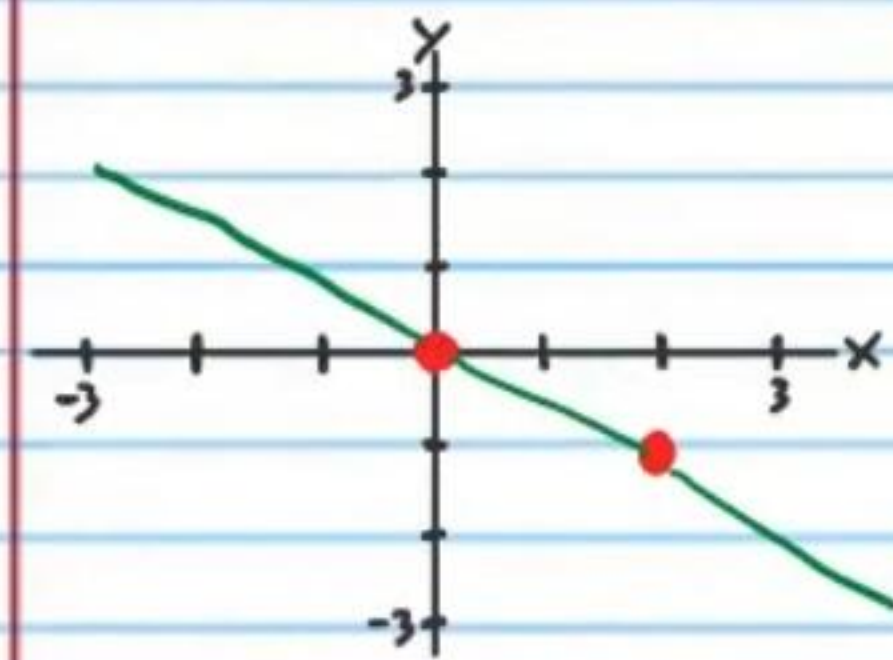
$y = 0$
↓
 $x = 0 - 3$
 $x = -3$
(-3,0)

2nd Point

$y = 1$
↓
 $x = 1 - 3$
 $x = -2$
(-2,1)

④ $X/-2 = Y$

X	Y
0	0
2	-1



1st Point

$$X=0$$



$$\frac{0}{-2} = Y$$

$$0 = Y$$

$$(0,0)$$

2nd Point

$$X=2$$



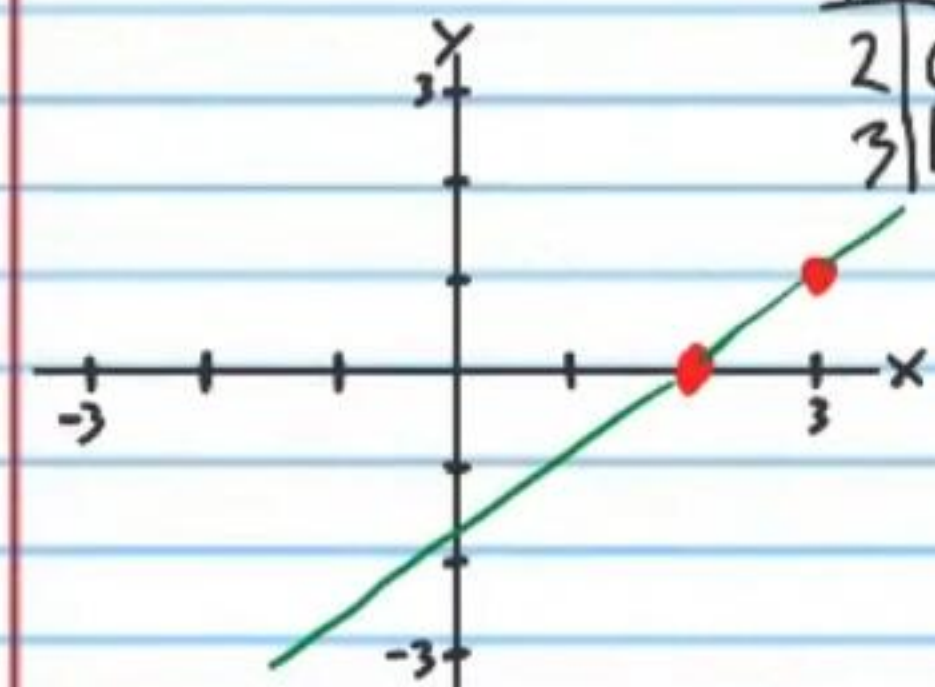
$$\frac{2}{-2} = Y$$

$$-1 = Y$$

$$(2,-1)$$

⑤ $X = Y + 2$

X	Y
2	0
3	1



1st Point

$$Y=0$$



$$X = 0 + 2$$

$$X = 2$$

$$(2,0)$$

2nd Point

$$Y=1$$



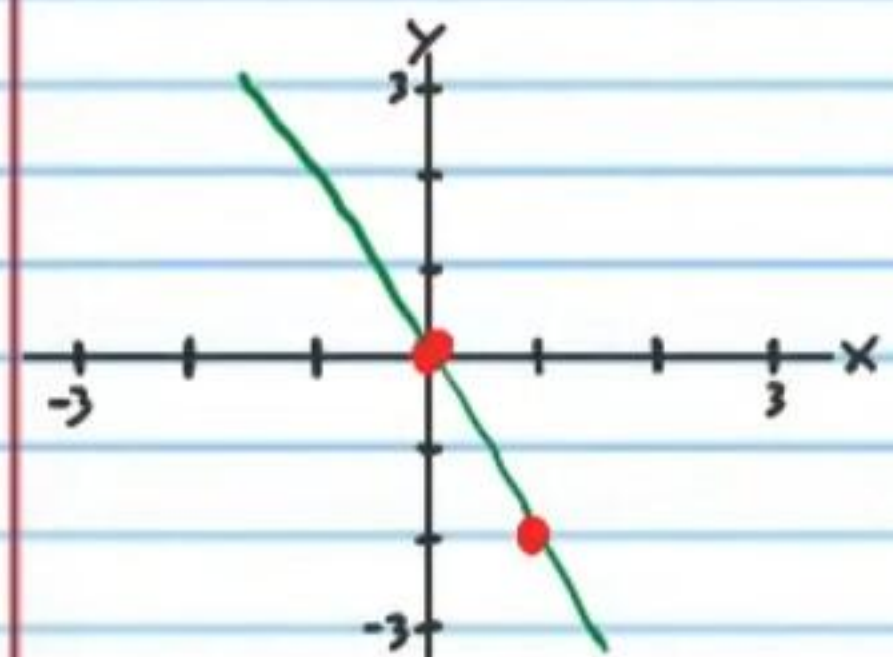
$$X = 1 + 2$$

$$X = 3$$

$$(3,1)$$

⑥ $Y = -2X$

X	Y
0	0
1	-2



1st Point

$$X=0$$



$$Y = -2(0)$$

$$Y = 0$$

$$(0,0)$$

2nd Point

$$X=1$$



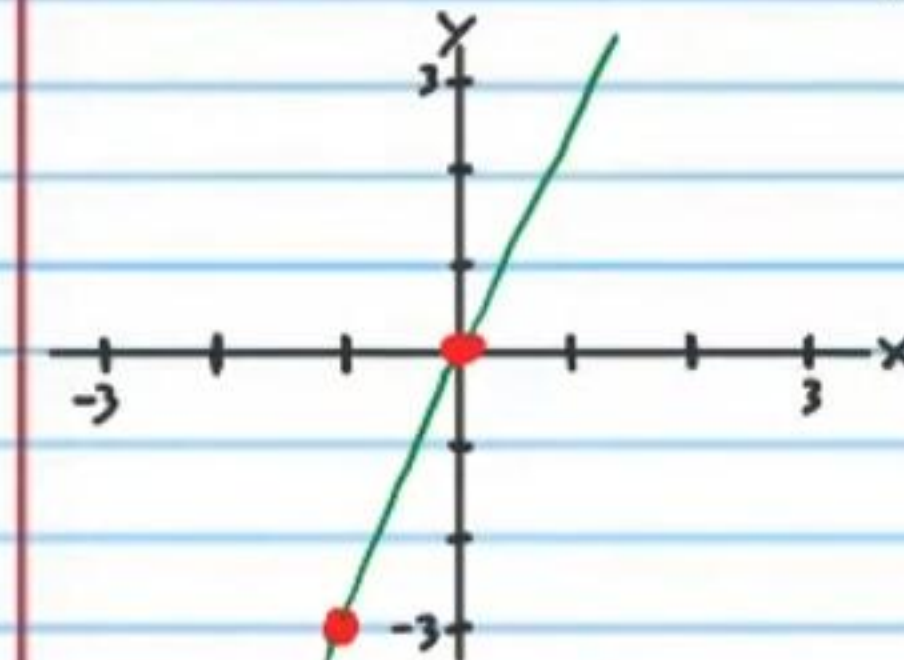
$$Y = -2(1)$$

$$Y = -2$$

$$(1,-2)$$

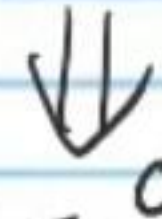
⑦ $X = Y/3$

X	Y
0	0
-1	-3



1st Point

$$Y=0$$



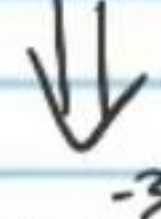
$$X = 0/3$$

$$X = 0$$

$$(X,0)$$

2nd Point

$$Y=-3$$



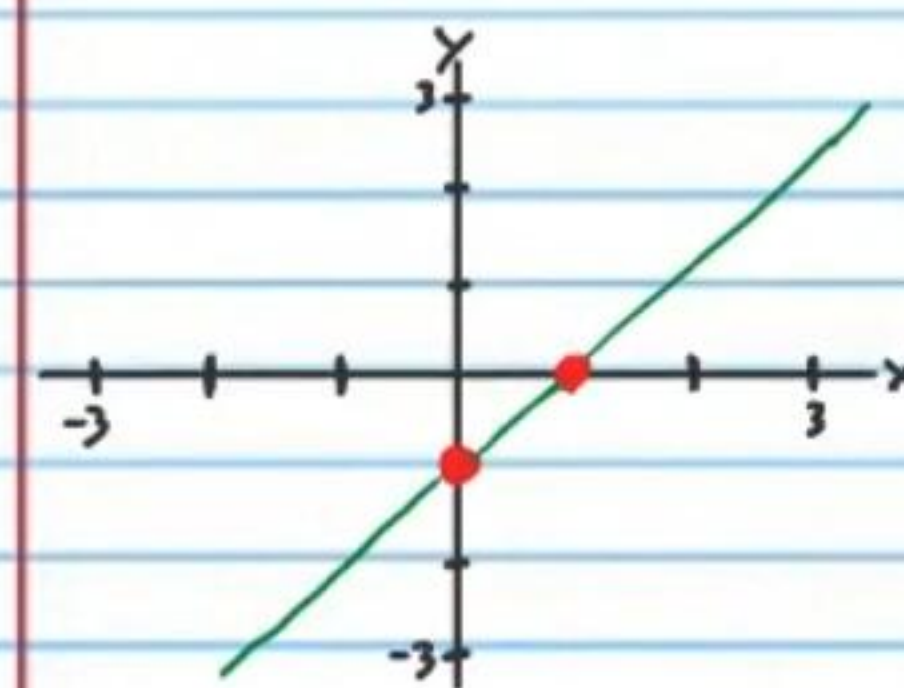
$$X = -3/3$$

$$X = -1$$

$$(-1,-3)$$

⑧ $Y = X - 1$

X	Y
0	-1
1	0



1st Point

$$X=0$$



$$Y = 0 - 1$$

$$Y = -1$$

$$(0,-1)$$

2nd Point

$$X=1$$



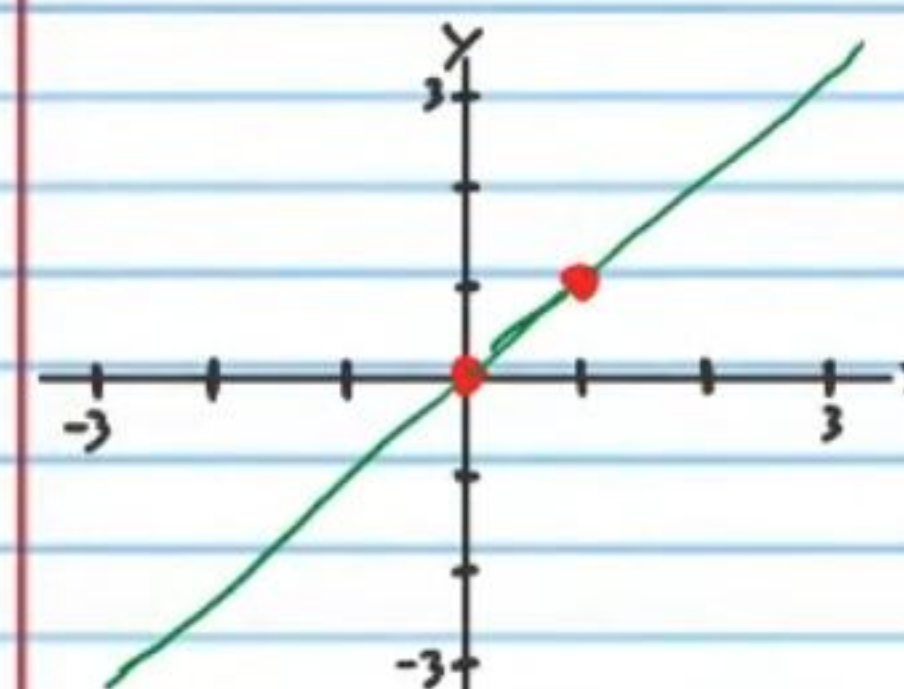
$$Y = 1 - 1$$

$$Y = 0$$

$$(1,0)$$

* ⑨ $Y = X$

X	Y
0	0
1	1



1st Point

$$X=0$$



$$Y=0$$

$$(0,0)$$

2nd Point

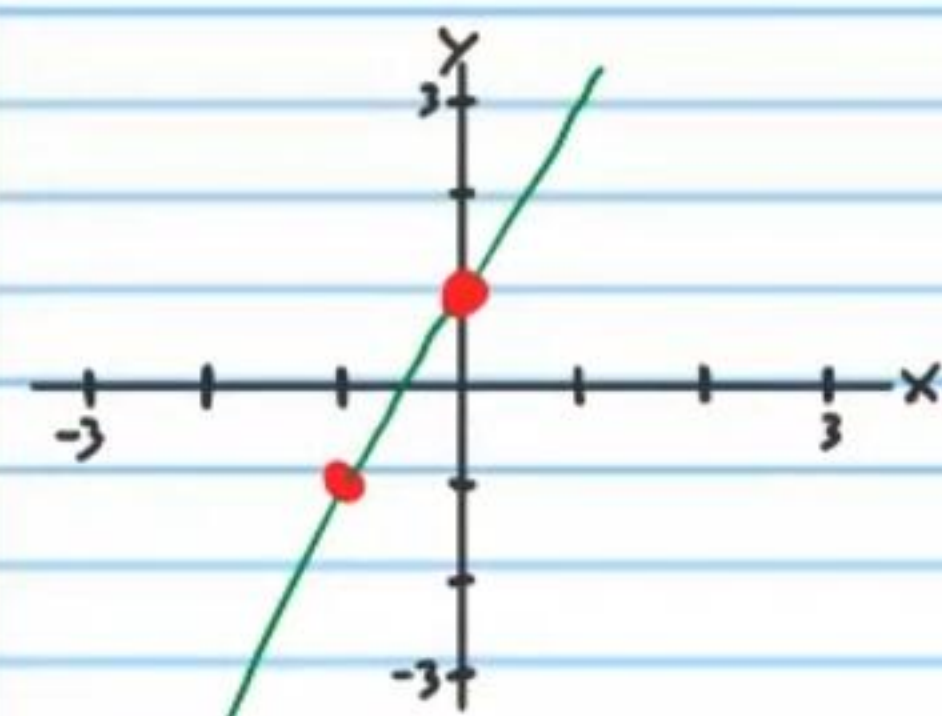
$$X=1$$



$$Y=1$$

$$(1,1)$$

* ⑩ $y = 2x + 1$

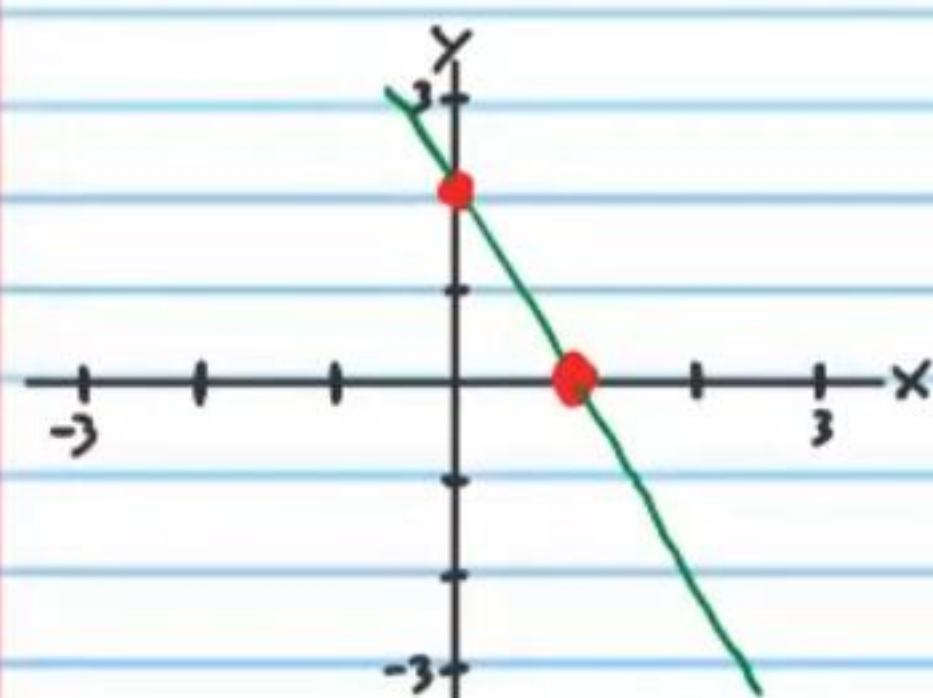


X	Y
0	1
-1	-1

1st Point 2nd Point

$X = 0$	$X = -1$
↓	↓
$y = 2(0) + 1$	$y = 2(-1) + 1$
$y = 0 + 1$	$y = -2 + 1$
$y = 1$	$y = -1$
$(0, 1)$	$(-1, -1)$

⑪ $x = 1 - \frac{y}{2}$

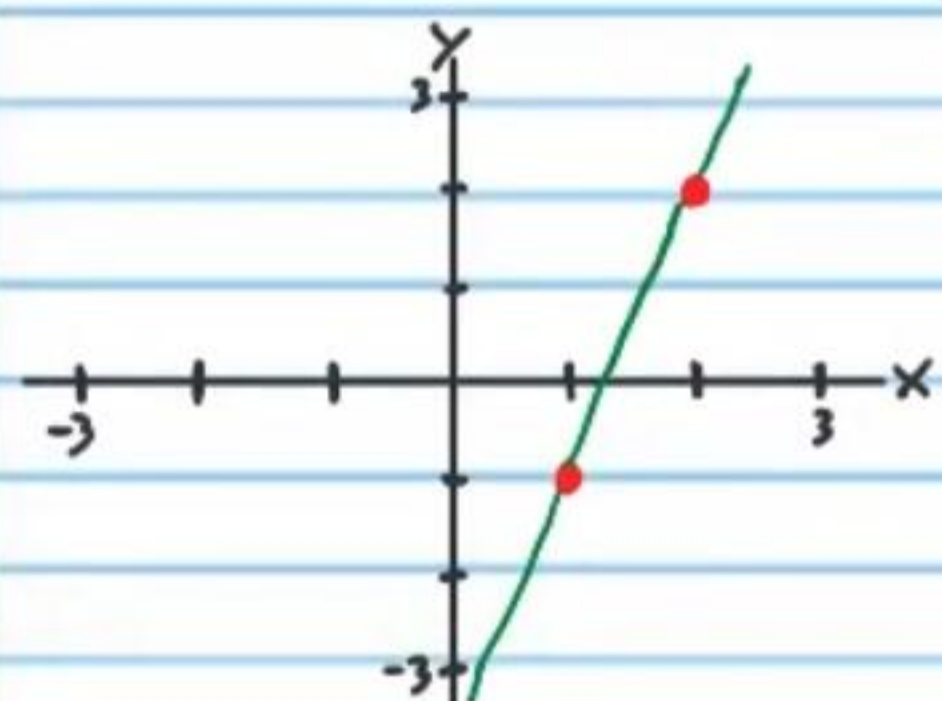


X	Y
1	0
0	2

1st Point 2nd Point

$Y = 0$	$Y = 2$
↓	↓
$X = 1 - \frac{0}{2}$	$X = 1 - \frac{2}{2}$
$X = 1 - 0$	$X = 1 - 1$
$X = 1$	$X = 0$
$(1, 0)$	$(0, 2)$

⑫ $-4 + 3x = y$

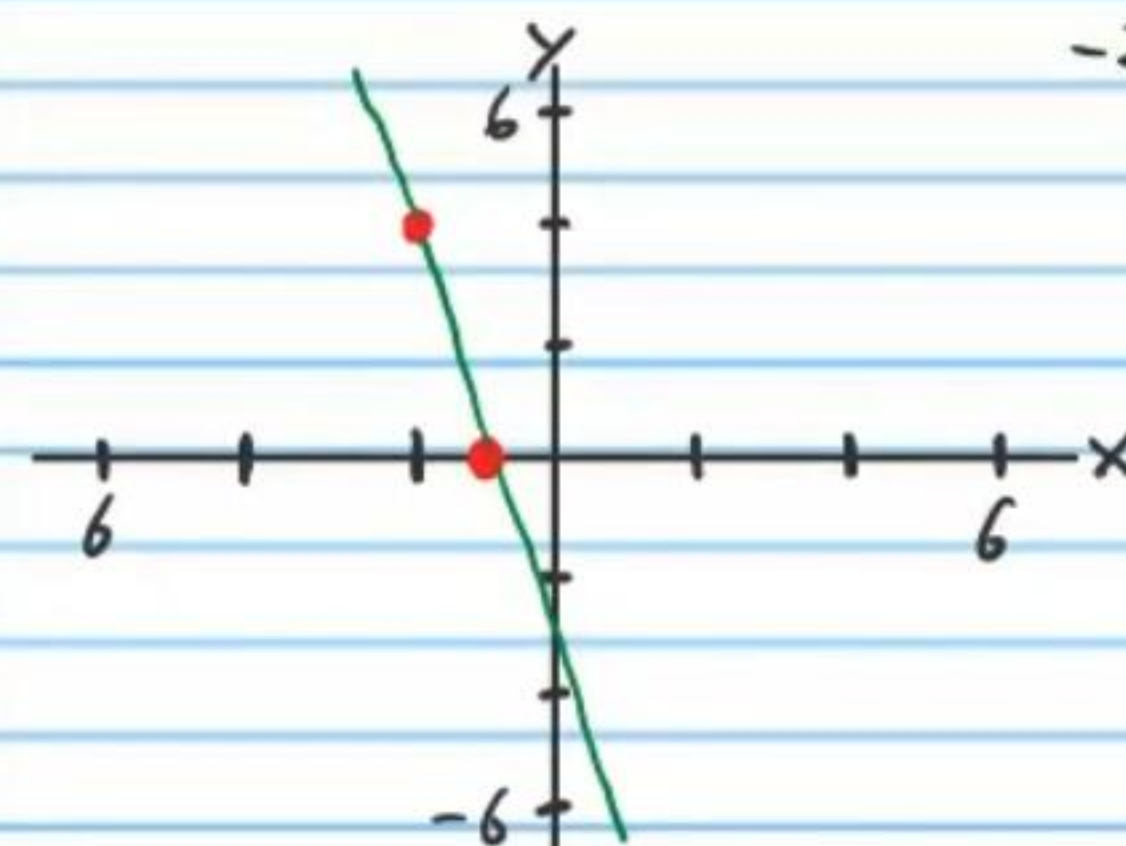


X	Y
1	-1
2	2

1st Point 2nd Point

$X = 1$	$X = 2$
↓	↓
$-4 + 3(1) = y$	$-4 + 3(2) = y$
$-4 + 3 = y$	$-4 + 6 = y$
$-1 = y$	$2 = y$
$(1, -1)$	$(2, 2)$

⑬ $x = -\frac{y}{4} - 1$

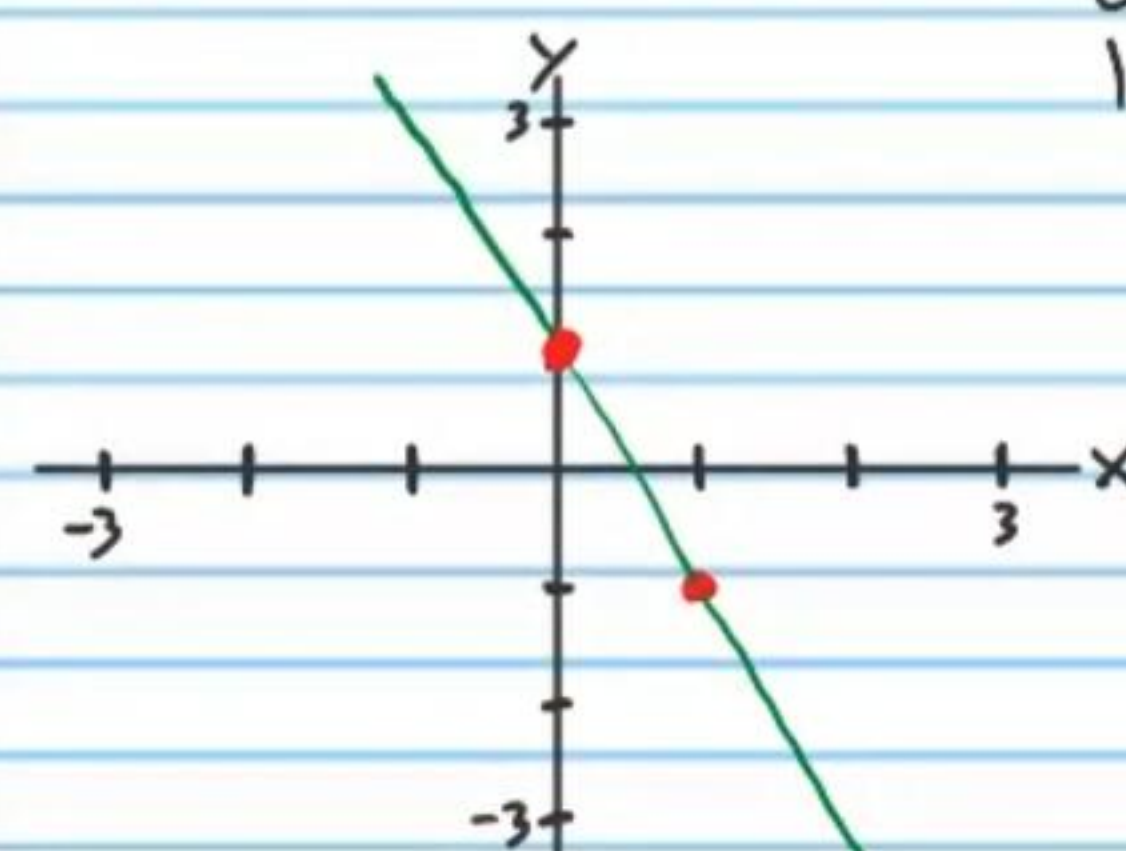


X	Y
-1	0
-2	4

1st Point 2nd Point

$Y = 0$	$Y = 4$
↓	↓
$X = -\frac{0}{4} - 1$	$X = -\frac{4}{4} - 1$
$X = -0 - 1$	$X = -1 - 1$
$X = 0 - 1$	$X = -2$
$X = -1$	$(-2, 4)$
$(-1, 0)$	

⑭ $y = -2x + 1$

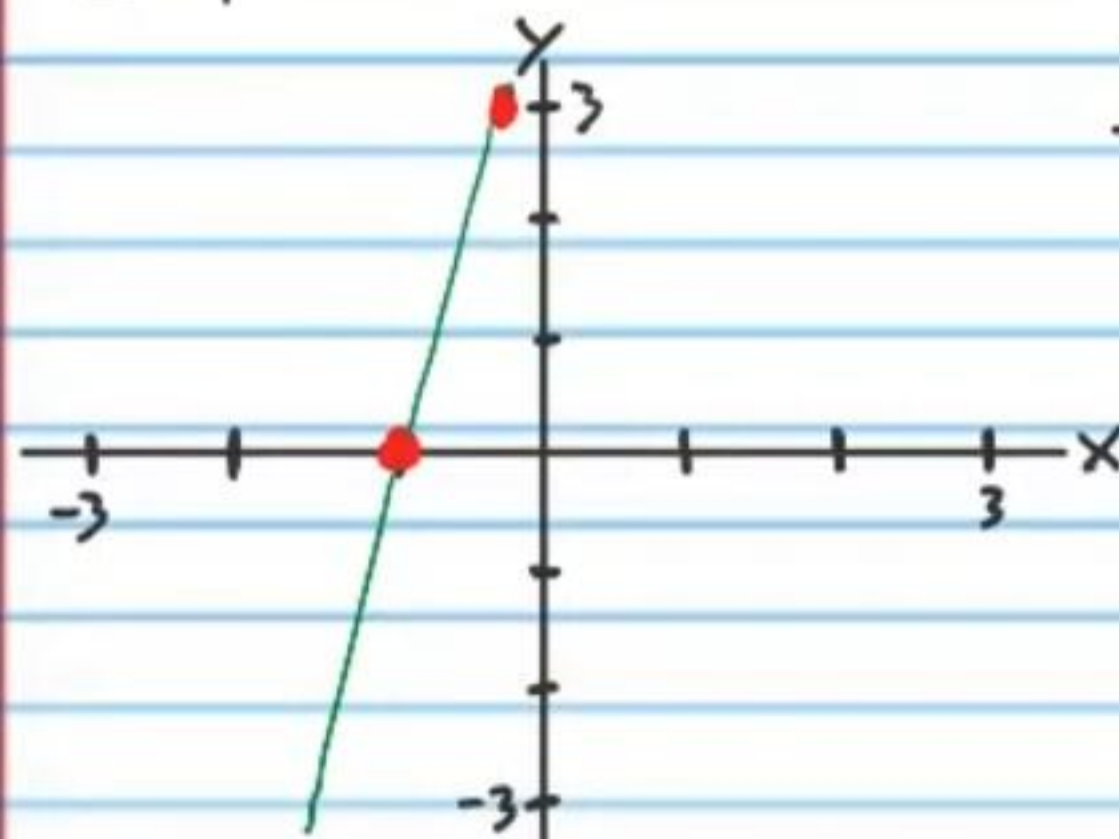


X	Y
0	1
1	-1

1st Point 2nd Point

$X = 0$	$X = 1$
↓	↓
$Y = -2(0) + 1$	$Y = -2(1) + 1$
$Y = 1$	$Y = -2 + 1$
$Y = 1$	$Y = -1$
$(0, 1)$	$(1, -1)$

⑮ $\frac{y}{4} - 1 = x$

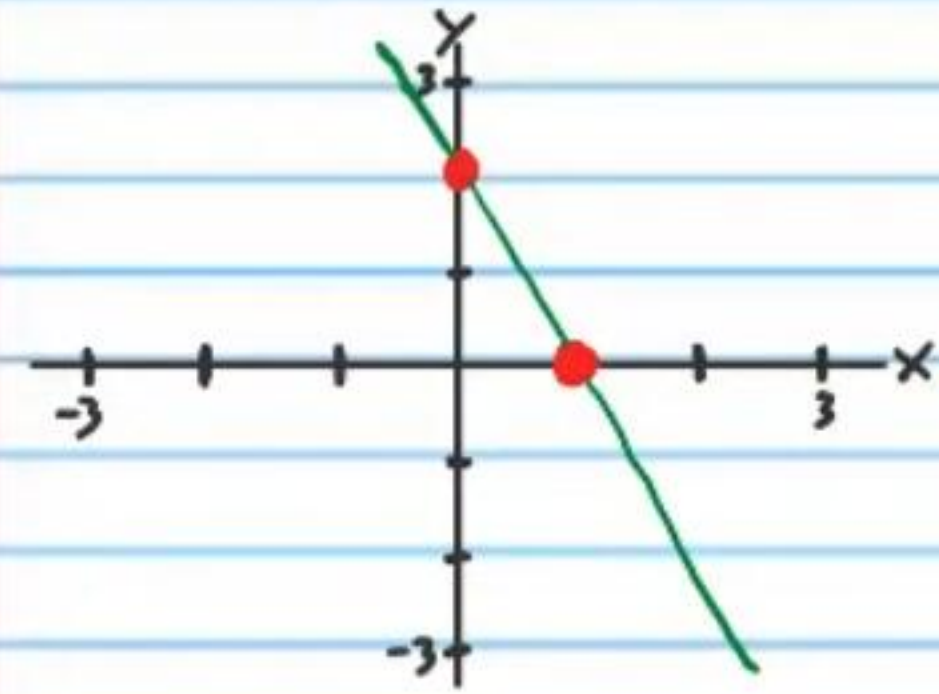


X	Y
-1	0
-1/4	3

1st Point 2nd Point

$Y = 0$	$Y = 3$
↓	↓
$\frac{0}{4} - 1 = x$	$\frac{3}{4} - 1 = x$
$0 - 1 = x$	$\frac{3}{4} - \frac{4}{4} = x$
$-1 = x$	$-\frac{1}{4} = x$
$(-1, 0)$	$(-\frac{1}{4}, 3)$

⑩ $2 - 2x = y$



x	y
0	2
1	0

1st Point

$x = 0$



$2 - 2(0) = y$

$2 - 0 = y$

$2 = y$

$(0, 2)$

2nd Point

$x = 1$



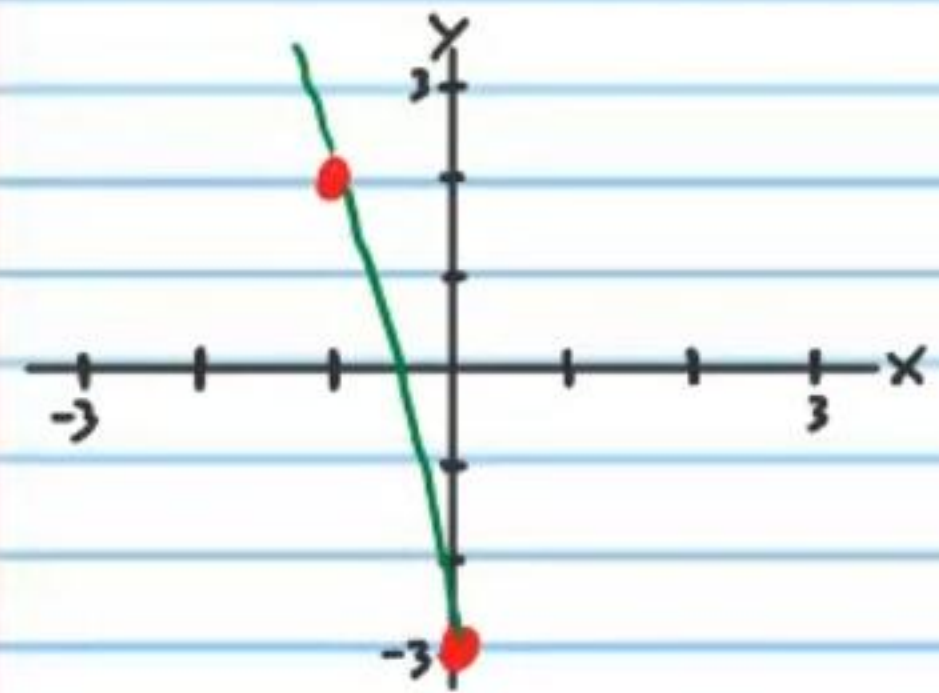
$2 - 2(1) = y$

$2 - 2 = y$

$0 = y$

$(1, 0)$

⑪ $-5x - 3 = y$



x	y
0	-3
-1	2

1st point

$x = 0$



$-5(0) - 3 = y$

$0 - 3 = y$

$-3 = y$

$(0, -3)$

2nd point

$x = -1$



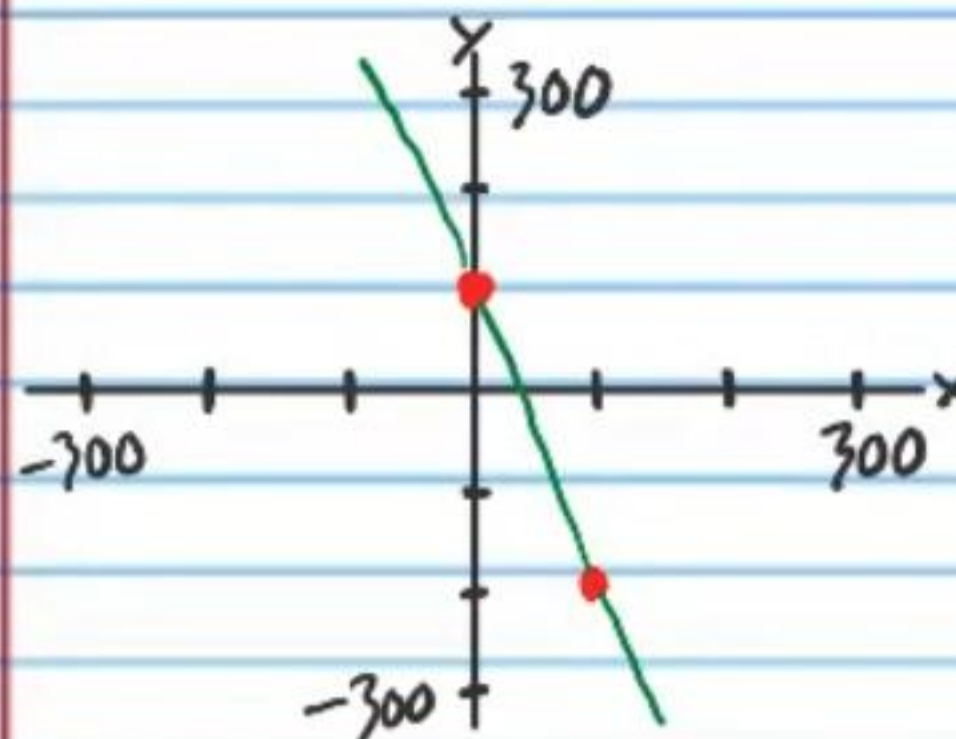
$-5(-1) - 3 = y$

$5 - 3 = y$

$2 = y$

$(-1, 2)$

* ⑫ $y = 100 - 3x$



x	y
0	100
100	-200

1st Point

$x = 0$



$y = 100 - 3(0)$

$y = 100 - 0$

$y = 100$

$(0, 100)$

2nd Point

$x = 100$



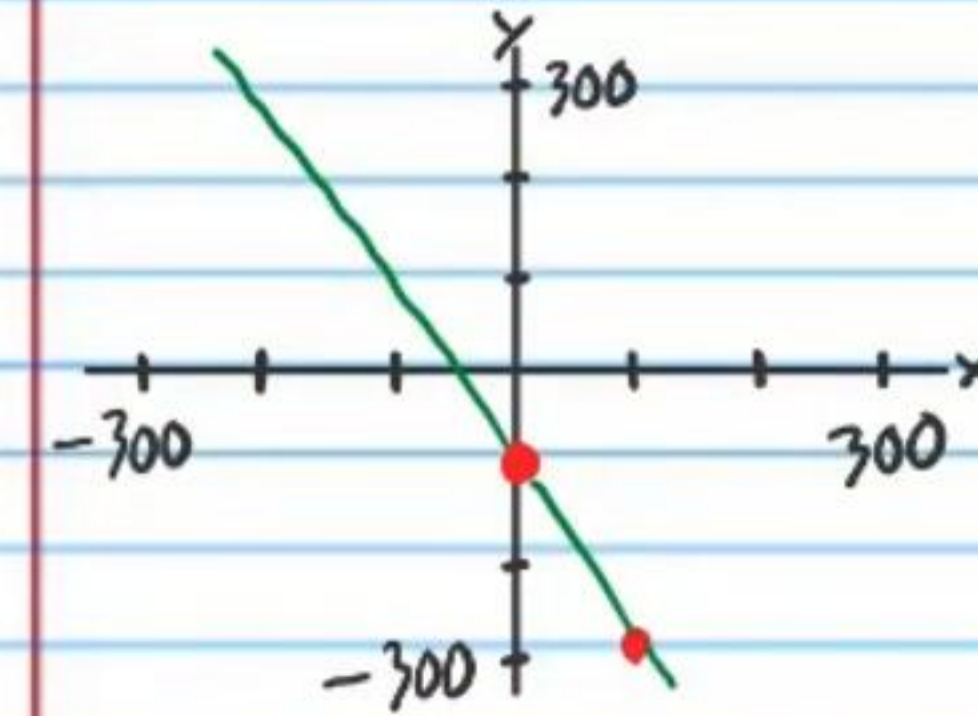
$y = 100 - 3(100)$

$y = 100 - 300$

$y = -200$

$(100, -200)$

⑬ $y = -2x - 100$



x	y
0	-100
100	-300

1st Point

$x = 0$



$y = -2(0) - 100$

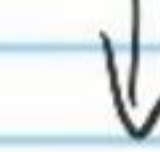
$y = 0 - 100$

$y = -100$

$(0, -100)$

2nd Point

$x = 100$



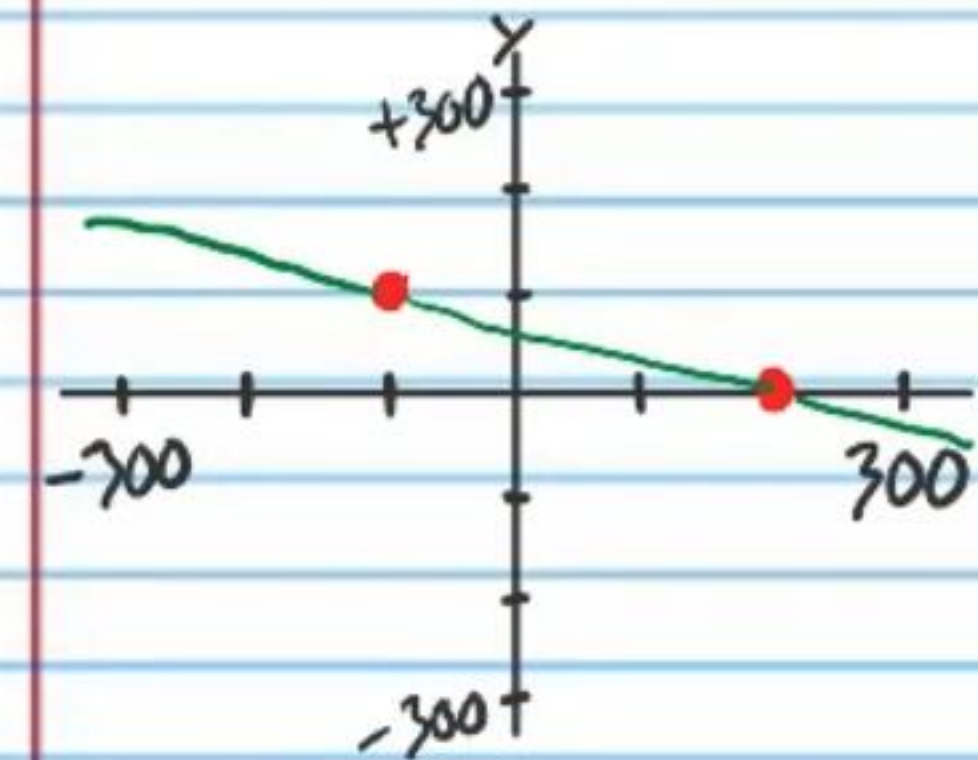
$y = -2(100) - 100$

$y = -200 - 100$

$y = -300$

$(100, -300)$

⑭ $x = -3y + 200$



x	y
200	0
-100	100

1st Point

$y = 0$



$x = -3(0) + 200$

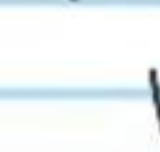
$x = 0 + 200$

$x = 200$

$(200, 0)$

2nd Point

$y = 100$



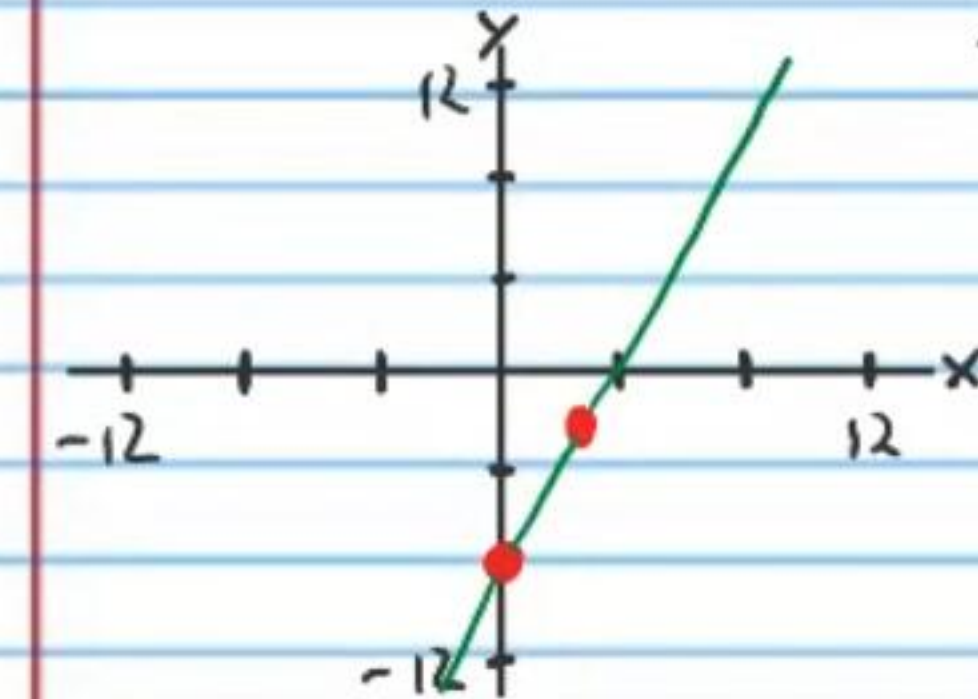
$x = -3(100) + 200$

$x = -300 + 200$

$x = -100$

$(-100, 100)$

⑮ $2x - 8 = y$



x	y
0	-8
3	-2

1st Point

$x = 0$



$2(0) - 8 = y$

$0 - 8 = y$

$-8 = y$

$(0, -8)$

2nd Point

$x = 3$



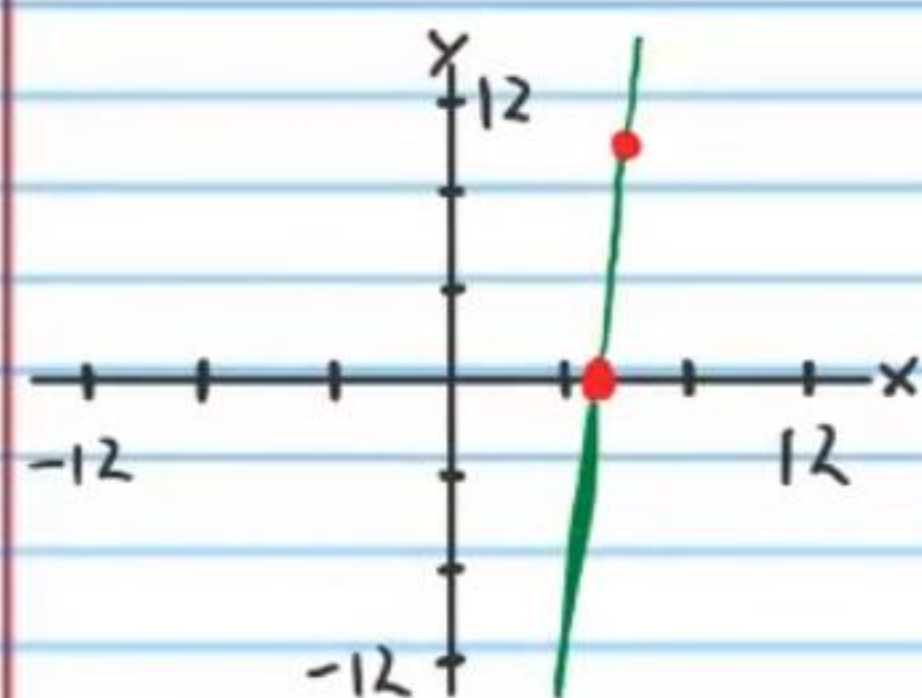
$2(3) - 8 = y$

$6 - 8 = y$

$-2 = y$

$(3, -2)$

②② $x = \frac{y}{10} + 5$



x	y
5	0
6	10

1st Point

$$y=0$$

$$\Downarrow$$

$$x = \frac{0}{10} + 5$$

$$x = 0 + 5$$

$$x = 5$$

(5, 0)

2nd Point

$$y=10$$

$$\Downarrow$$

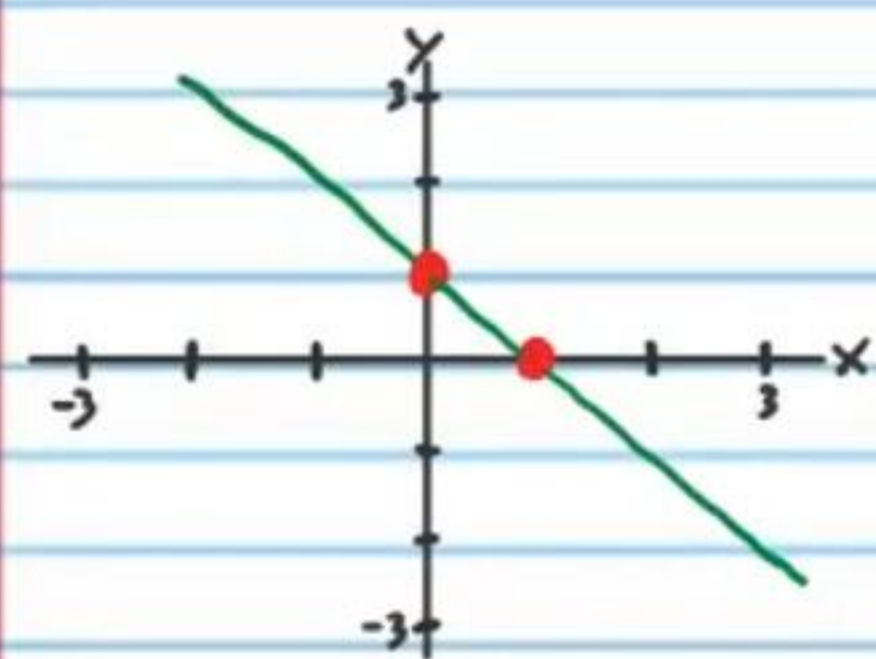
$$x = \frac{10}{10} + 5$$

$$x = 1 + 5$$

$$x = 6$$

(6, 10)

* ②③ $x + y = 1$



x	y
0	1
1	0

1st Point

$$x=0$$

$$\Downarrow$$

$$0 + y = 1$$

$$y = 1$$

(0, 1)

2nd Point

$$x=1$$

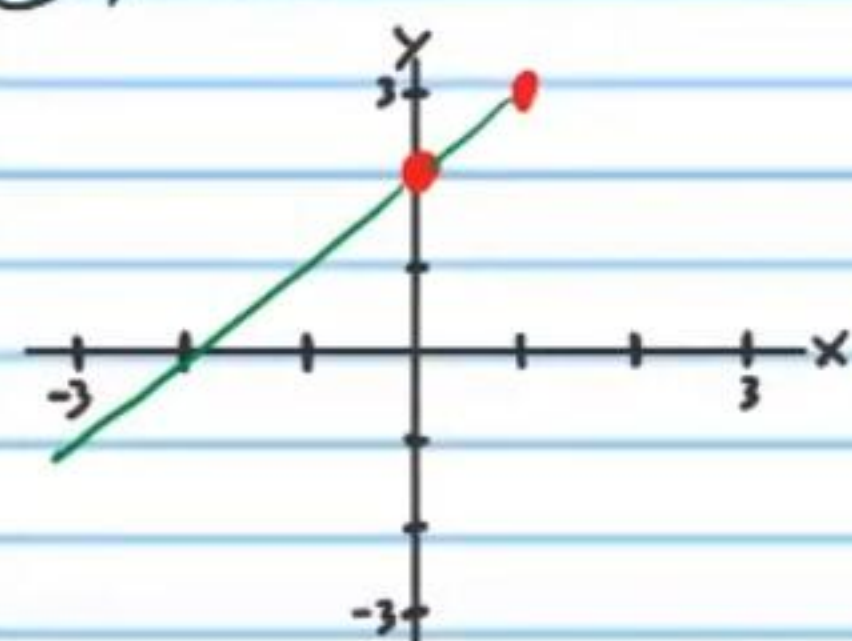
$$\Downarrow$$

$$1 + y = 1$$

$$y = 0$$

(1, 0)

②④ $y - x = 2$



x	y
0	2
1	3

1st Point

$$x=0$$

$$\Downarrow$$

$$y - 0 = 2$$

$$y = 2$$

(0, 2)

2nd Point

$$x=1$$

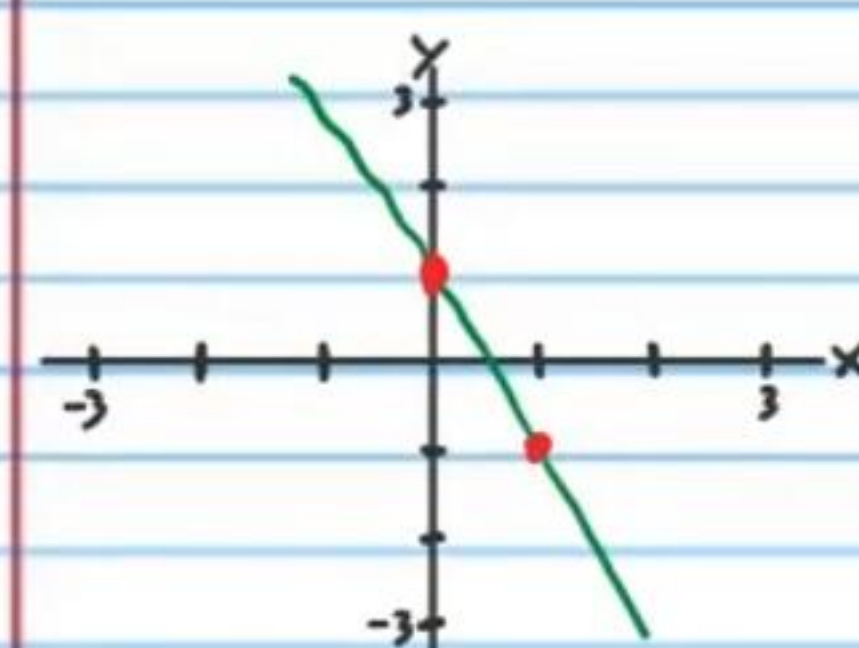
$$\Downarrow$$

$$y - 1 = 2$$

$$y = 3$$

(1, 3)

②⑤ $2x + y = 1$



x	y
0	1
1	-1

1st Point

$$x=0$$

$$\Downarrow$$

$$2(0) + y = 1$$

$$0 + y = 1$$

$$y = 1$$

(0, 1)

2nd Point

$$x=1$$

$$\Downarrow$$

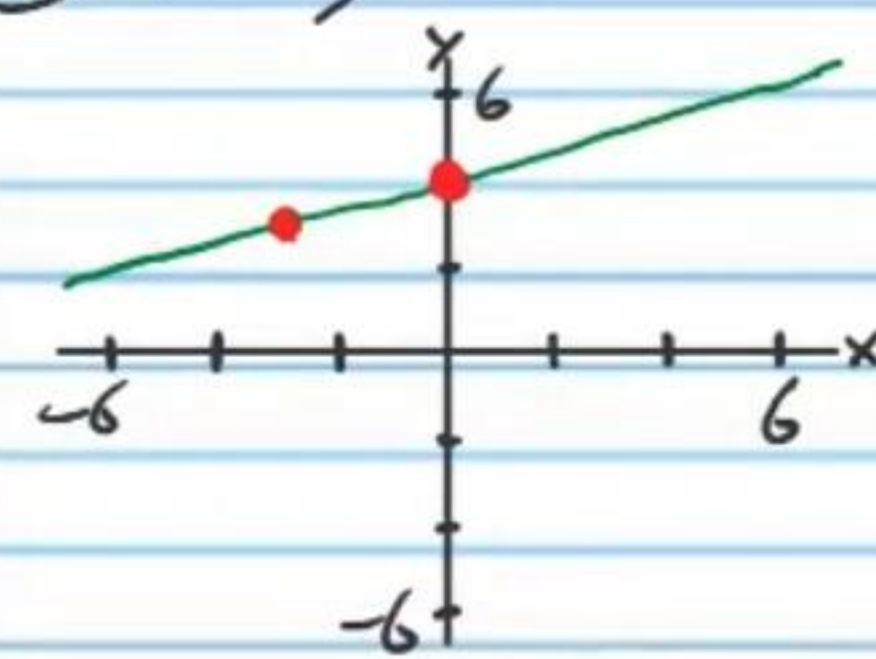
$$2(1) + y = 1$$

$$2 + y = 1$$

$$y = -1$$

(1, -1)

②⑥ $y - \frac{x}{3} = 4$



x	y
0	4
-3	3

1st Point

$$x=0$$

$$\Downarrow$$

$$y - \frac{0}{3} = 4$$

$$y - 0 = 4$$

$$y = 4$$

(0, 4)

2nd Point

$$x=-3$$

$$\Downarrow$$

$$y - \left(\frac{-3}{3}\right) = 4$$

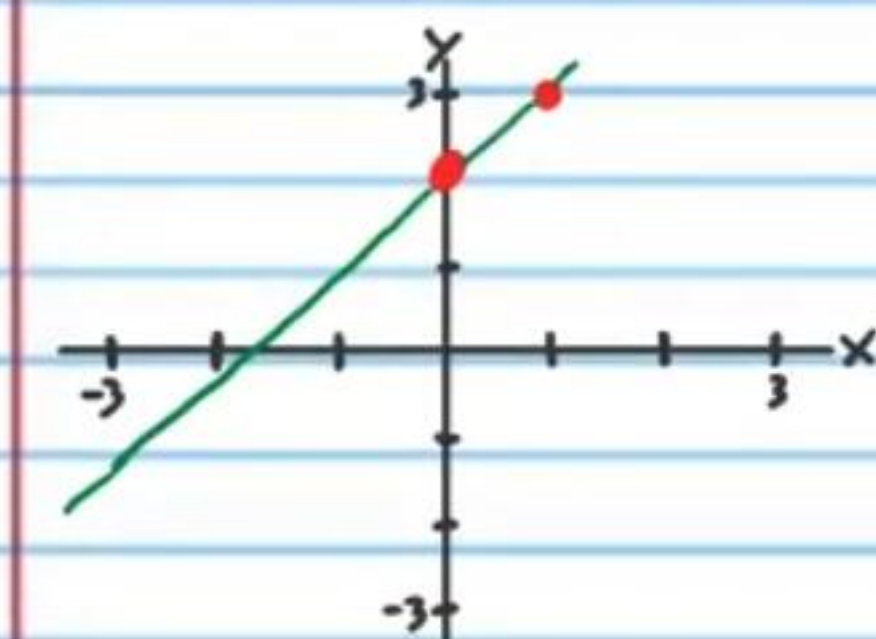
$$y + \frac{3}{3} = 4$$

$$y + 1 = 4$$

$$y = 3$$

(-3, 3)

②⑦ $-x + y = 2$



x	y
0	2
1	3

1st Point

$$x=0$$

$$\Downarrow$$

$$-0 + y = 2$$

$$y = 2$$

(0, 2)

2nd Point

$$x=1$$

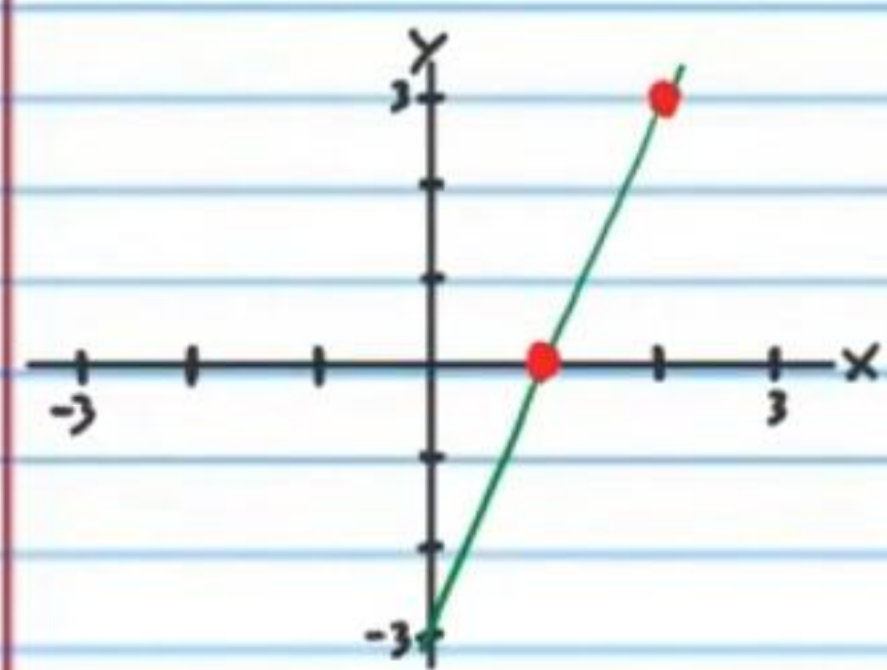
$$\Downarrow$$

$$-1 + y = 2$$

$$y = 3$$

(1, 3)

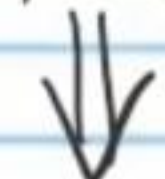
②⑧ $\frac{x}{3} + x = 1$



x	y
1	0
2	3

1st Point

$y = 0$



$\frac{0}{3} + x = 1$

$0 + x = 1$

$x = 1$

$(1, 0)$

2nd Point

$y = 3$



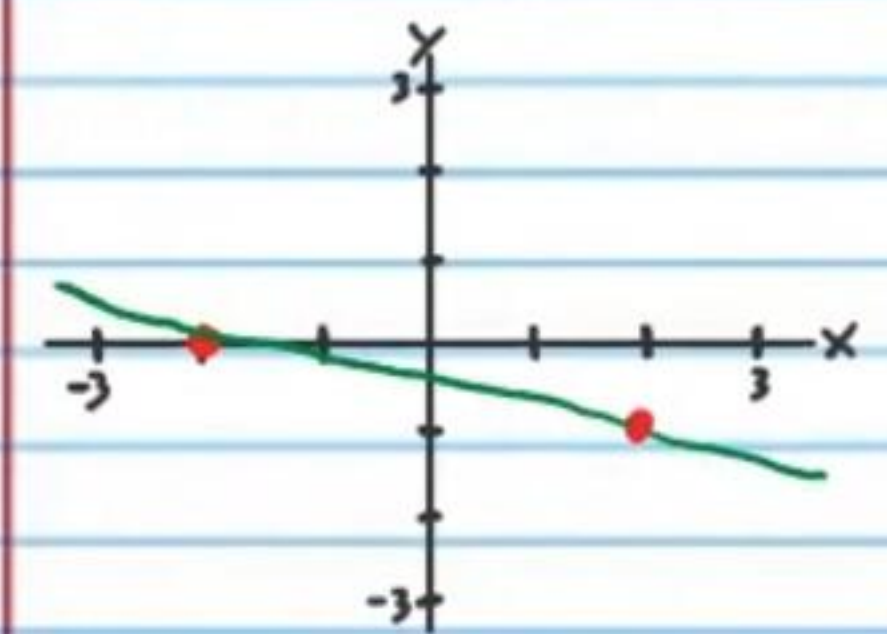
$\frac{3}{3} + x = 1$

$-1 + x = 1$

$x = 2$

$(2, 3)$

②⑨ $-x - 4y = 2$



x	y
-2	0
2	-1

1st Point

$y = 0$



$-x - 4(0) = 2$

$-x - 0 = 2$

$-x = 2$

$-1x = 2$

$x = -2$

$(-2, 0)$

2nd Point

$y = -1$



$-x - 4(-1) = 2$

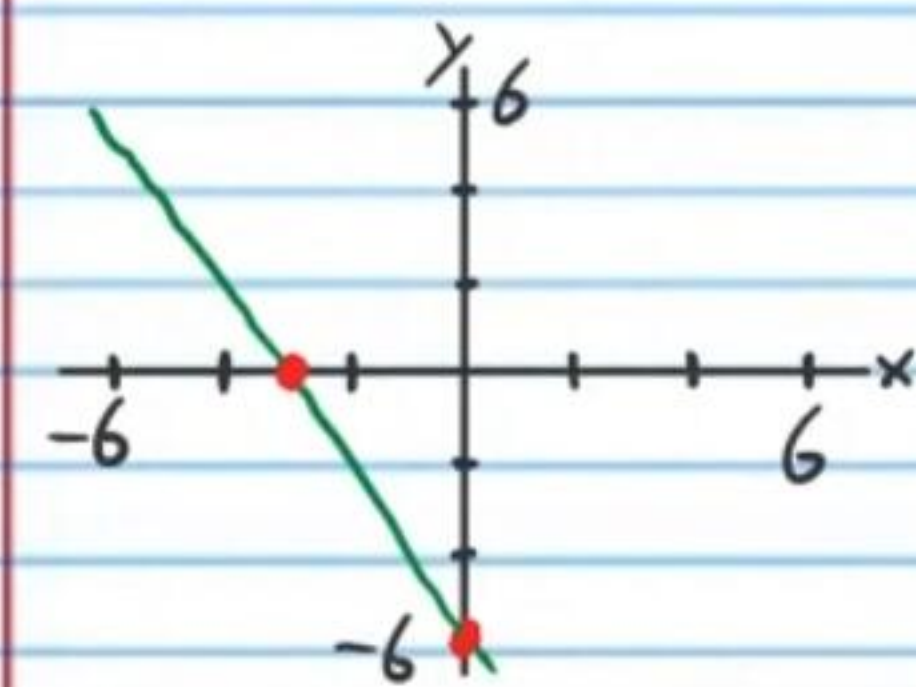
$-x + 4 = 2$

$-x = -2$

$x = 2$

$(2, -1)$

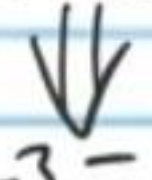
③⑩ $-\frac{y}{2} - 3 = x$



x	y
0	-6
-3	0

1st Point

$x = 0$



$-\frac{y}{2} - 3 = 0$

$-\frac{y}{2} = 3$

$y = -6$

$(0, -6)$

2nd Point

$y = 0$



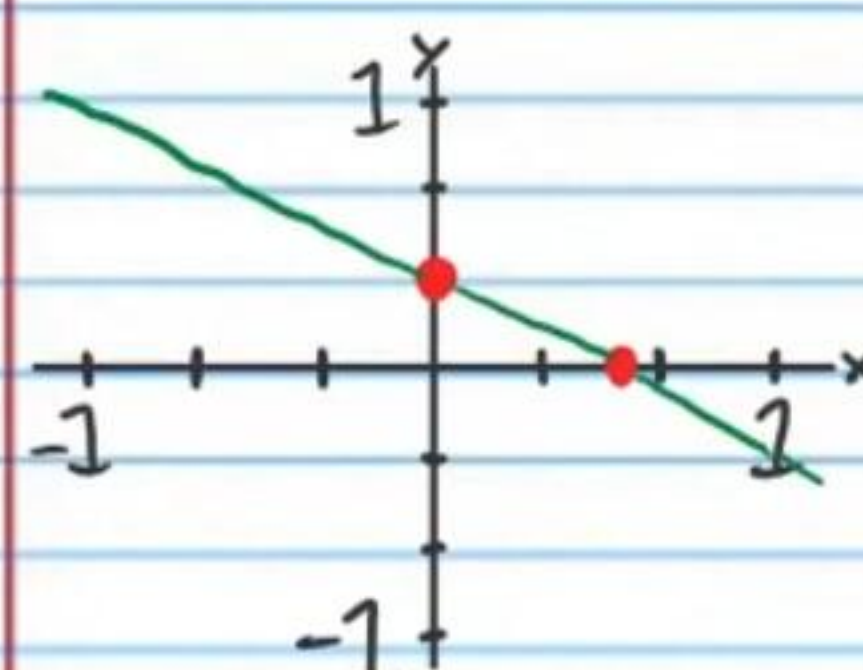
$-\frac{0}{2} - 3 = x$

$0 - 3 = x$

$-3 = x$

$(-3, 0)$

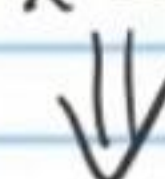
* ③⑪ $2x + 3y = 1$



x	y
0	1/3
1/2	0

1st Point

$x = 0$



$2(0) + 3y = 1$

$0 + 3y = 1$

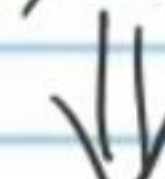
$3y = 1$

$y = \frac{1}{3}$

$(0, \frac{1}{3})$

2nd Point

$y = 0$



$2x + 3(0) = 1$

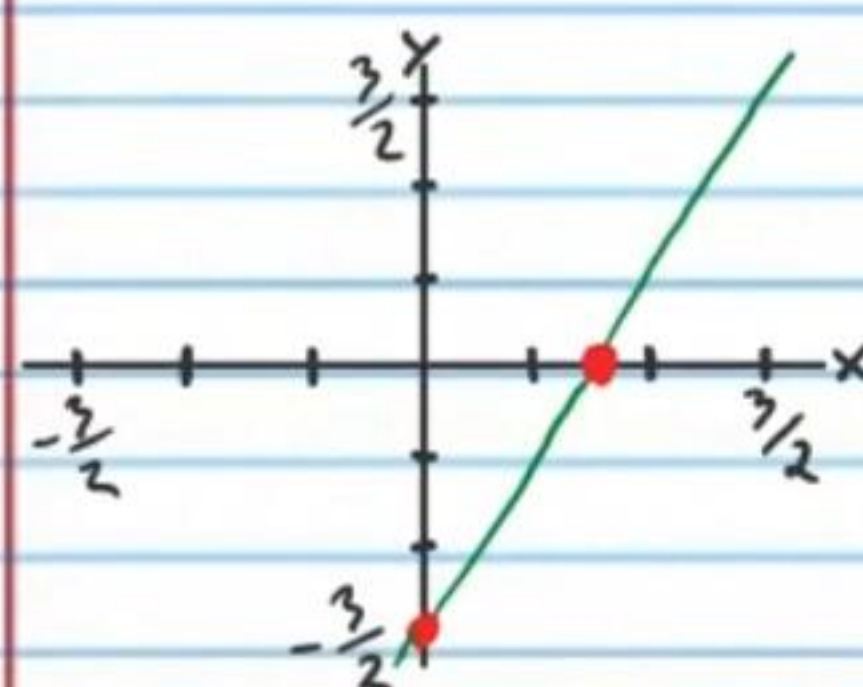
$2x + 0 = 1$

$2x = 1$

$x = \frac{1}{2}$

$(\frac{1}{2}, 0)$

③⑫ $-4x + 2y = -3$



x	y
0	-3/2
3/4	0

1st Point

$x = 0$



$-4(0) + 2y = -3$

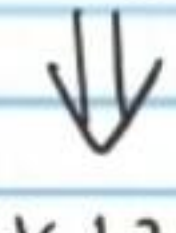
$0 + 2y = -3$

$y = -\frac{3}{2}$

$(0, -\frac{3}{2})$

2nd Point

$y = 0$



$-4x + 2(0) = -3$

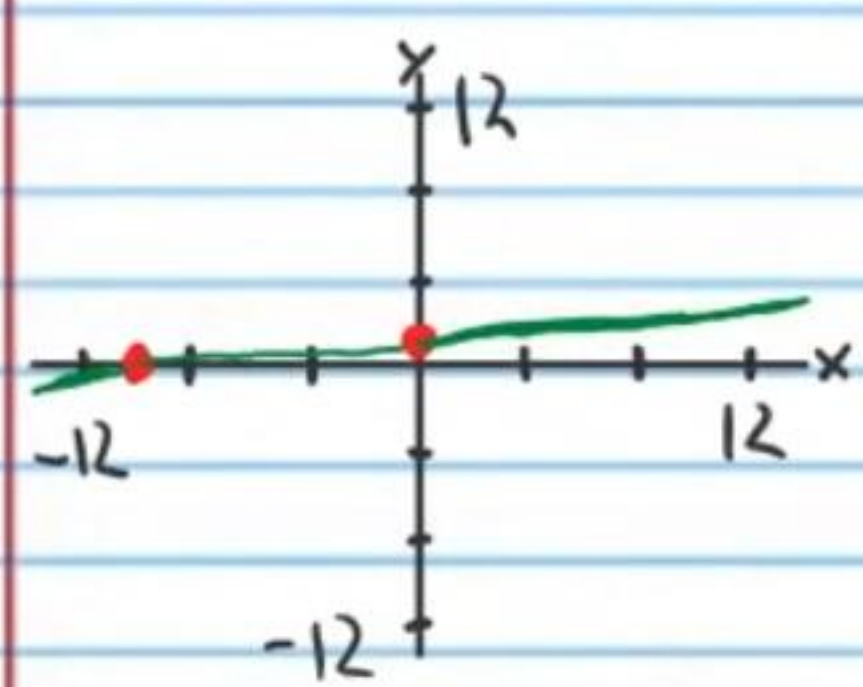
$-4x + 0 = -3$

$-4x = -3$

$x = \frac{3}{4}$

$(\frac{3}{4}, 0)$

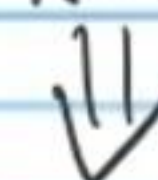
③⑬ $-\frac{x}{2} + 3y = 5$



x	y
0	5/3
-10	0

1st Point

$x = 0$



$-\frac{0}{2} + 3y = 5$

$y = \frac{5}{3}$

$(0, \frac{5}{3})$

$(0, \frac{5}{3})$

2nd Point

$y = 0$



$-\frac{x}{2} + 3(0) = 5$

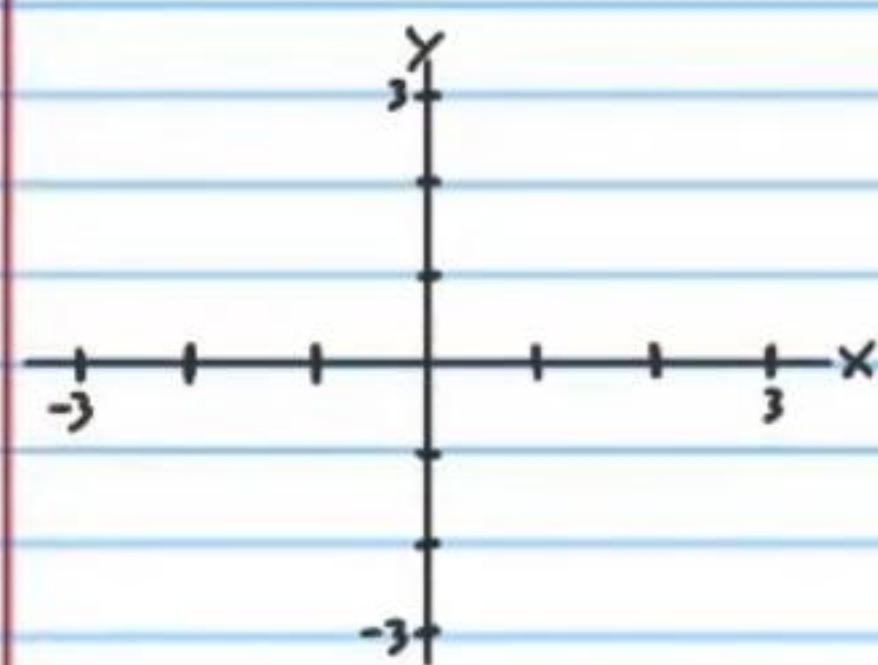
$-\frac{x}{2} + 0 = 5$

$-\frac{x}{2} = 5$

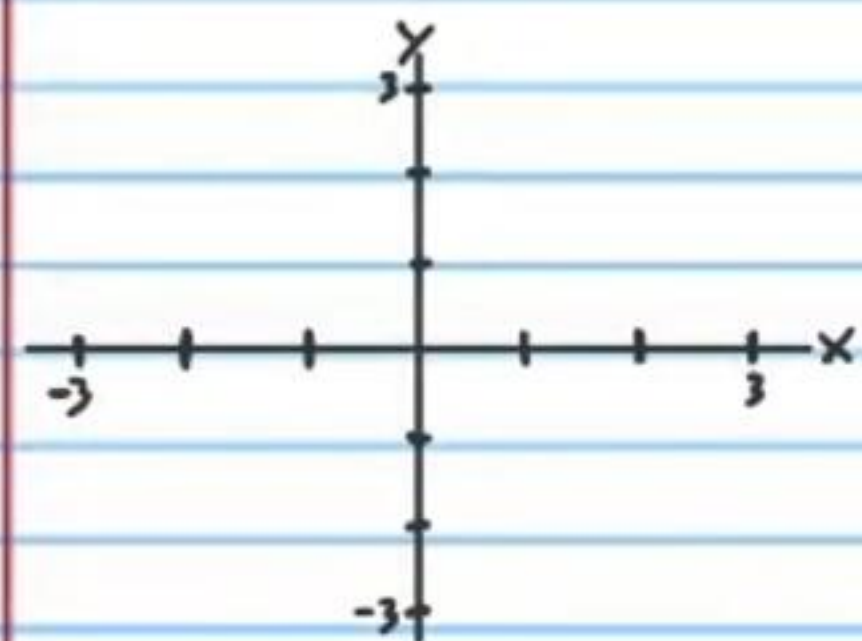
$x = -10$

$(-10, 0)$

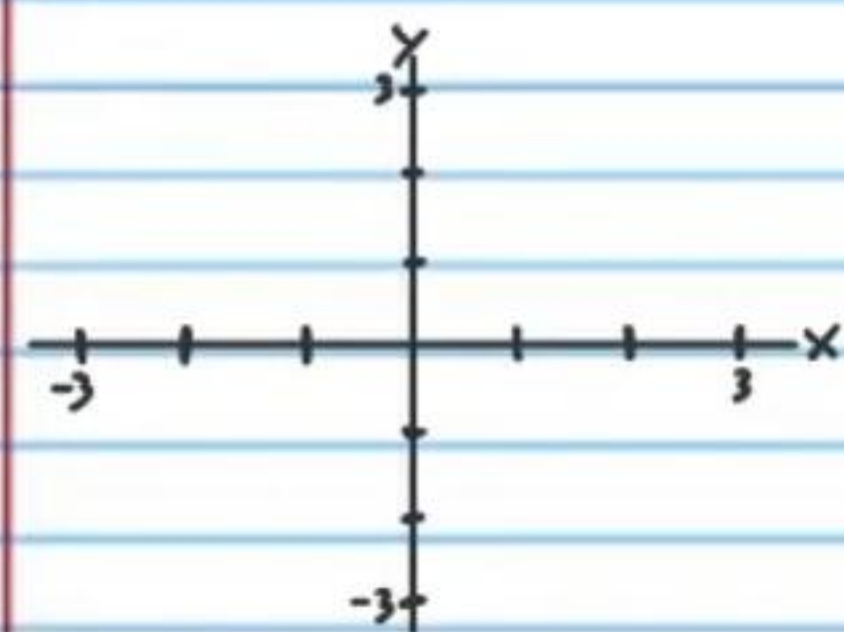
$$(34) 2x - \frac{y}{4} = -1$$



$$(35) \frac{x}{2} - \frac{y}{3} = 3$$



$$(36) -2 = -\frac{y}{2} - \frac{x}{4}$$



Graphing Linear Inequalities

$$(37) y \leq 4 + x$$

x	y
0	4
-2	2

1st Point

$$x = 0$$

$$\Downarrow$$

$$y \leq 4 + 0$$

$$y \leq 4$$

(0, 4)

2nd Point

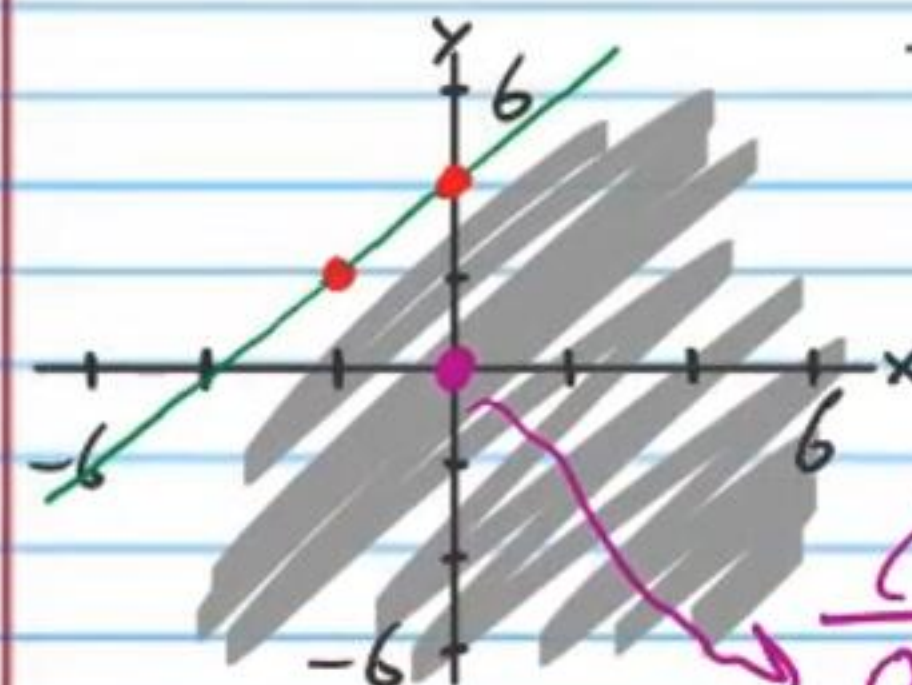
$$x = -2$$

$$\Downarrow$$

$$y \leq 4 + (-2)$$

$$y \leq 2$$

(-2, 2)



check

$$0 \leq 4 + 0$$

$$0 \leq 4$$

$$(38) x > -y$$

x	y
0	0
1	-1

1st Point

$$x = 0$$

$$\Downarrow$$

$$0 > -y$$

$$0 > y$$

(0, 0)

2nd Point

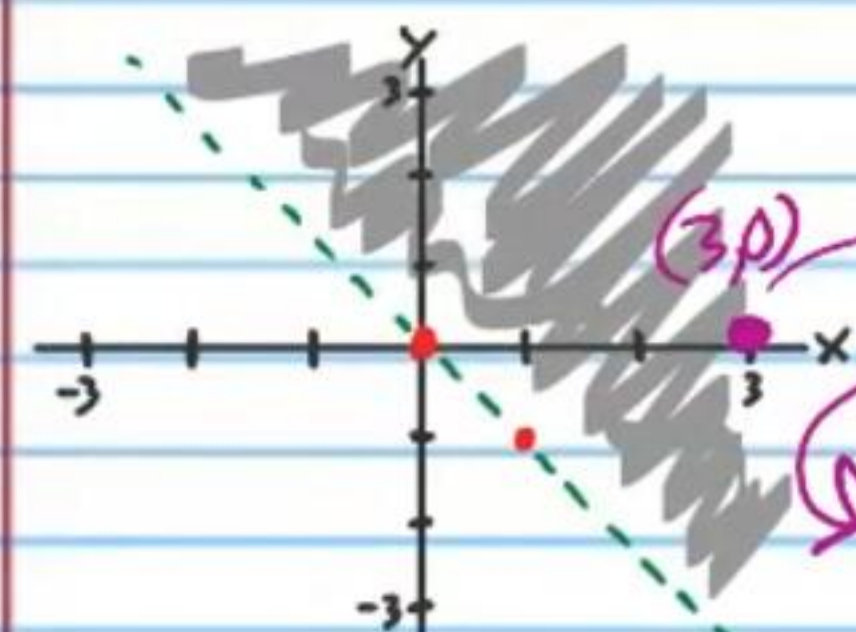
$$x = 1$$

$$\Downarrow$$

$$1 > -y$$

$$-1 > y$$

(1, -1)



check

$$3 > -0$$

$$3 > 0$$

$$(39) x \geq 2y - 1$$

x	y
-1	0
1	1

1st Point

$$y = 0$$

$$\Downarrow$$

$$x = 2(0) - 1$$

$$x = -1$$

(-1, 0)

2nd Point

$$y = 1$$

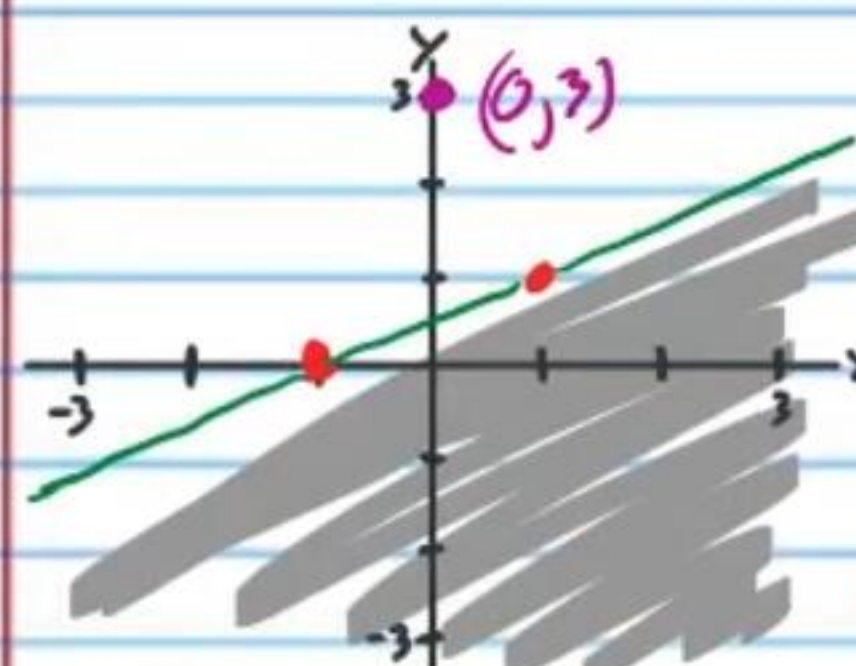
$$\Downarrow$$

$$x = 2(1) - 1$$

$$x = 2 - 1$$

$$x = 1$$

(1, 1)



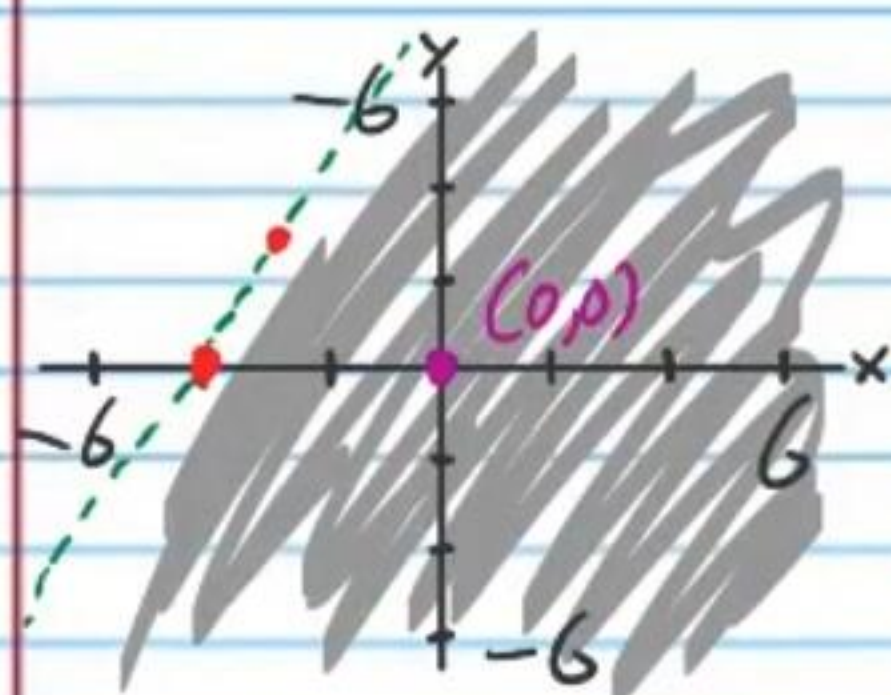
Test Point

$$0 \geq 2(3) - 1$$

$$0 \geq 6 - 1$$

$$0 \geq 5$$

④② $\frac{y}{3} - 4 < x$



x	y
-4	0
-3	3

Test Point
 $\frac{0}{3} - 4 < 0$
 $0 - 4 < 0$
 $-4 < 0$

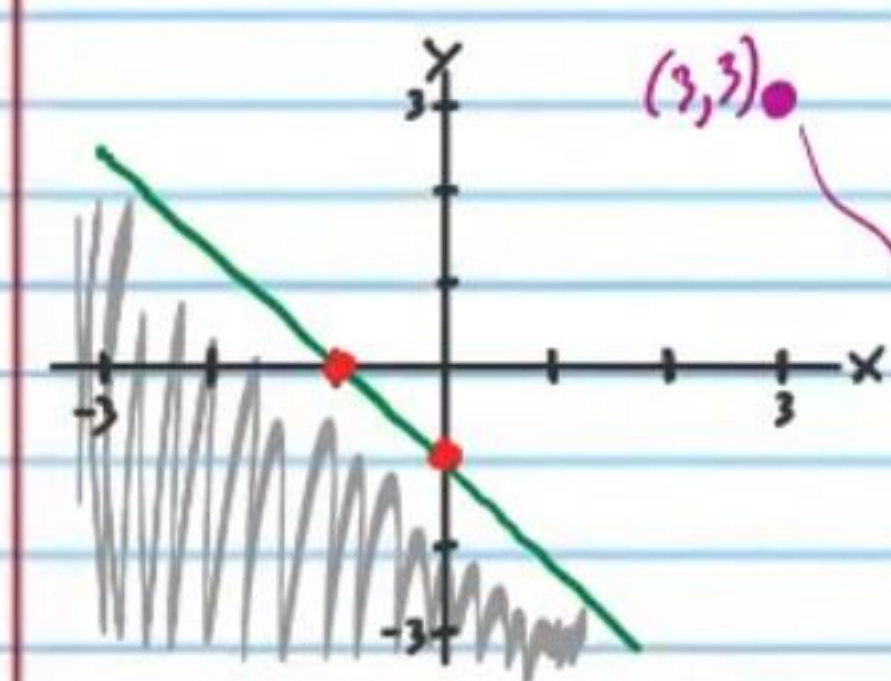
1st Point

$y = 0$
 $\Downarrow$
 $\frac{0}{3} - 4 = x$
 $-4 = x$
 $(-4, 0)$

2nd Point

$y = 3$
 $\Downarrow$
 $\frac{3}{3} - 4 = x$
 $1 - 4 = x$
 $-3 = x$
 $(-3, 3)$

④③ $x + y \leq -1$



x	y
0	-1
-1	0

Test Point
 $3 + 3 \leq -1$
 $6 \leq -1$

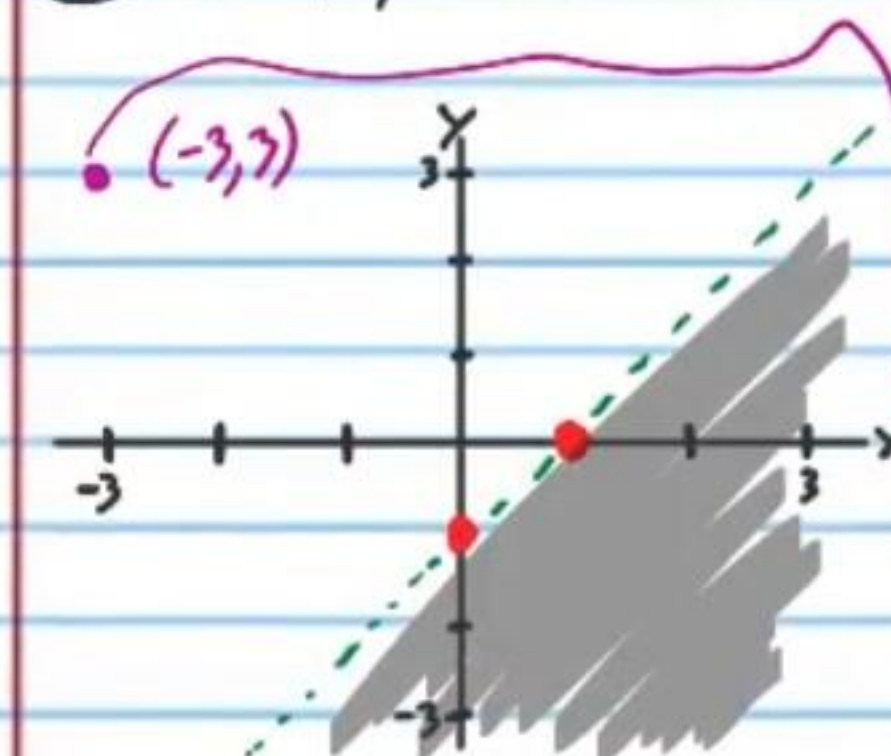
1st Point

$x = 0$
 $\Downarrow$
 $0 + y = -1$
 $y = -1$
 $(0, -1)$

2nd Point

$y = 0$
 $\Downarrow$
 $x + 0 = -1$
 $x = -1$
 $(-1, 0)$

④④ $1 + y < x$



x	y
1	0
0	-1

Test Point
 $1 + 3 < -3$
 $4 < -3$

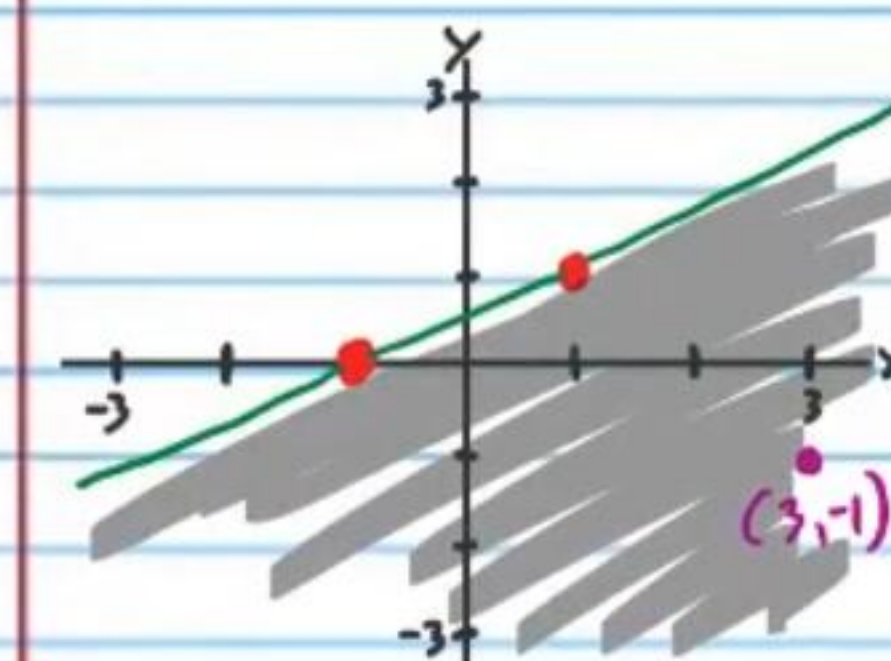
1st Point

$y = 0$
 $\Downarrow$
 $1 + 0 = x$
 $1 = x$
 $(1, 0)$

2nd Point

$y = -1$
 $\Downarrow$
 $1 + (-1) = x$
 $0 = x$
 $(0, -1)$

④⑤ $-1 + 2y \leq x$



x	y
-1	0
1	1

Test Point
 $-1 + 2(-1) \leq 3$
 $-1 - 2 \leq 3$
 $-3 \leq 3$

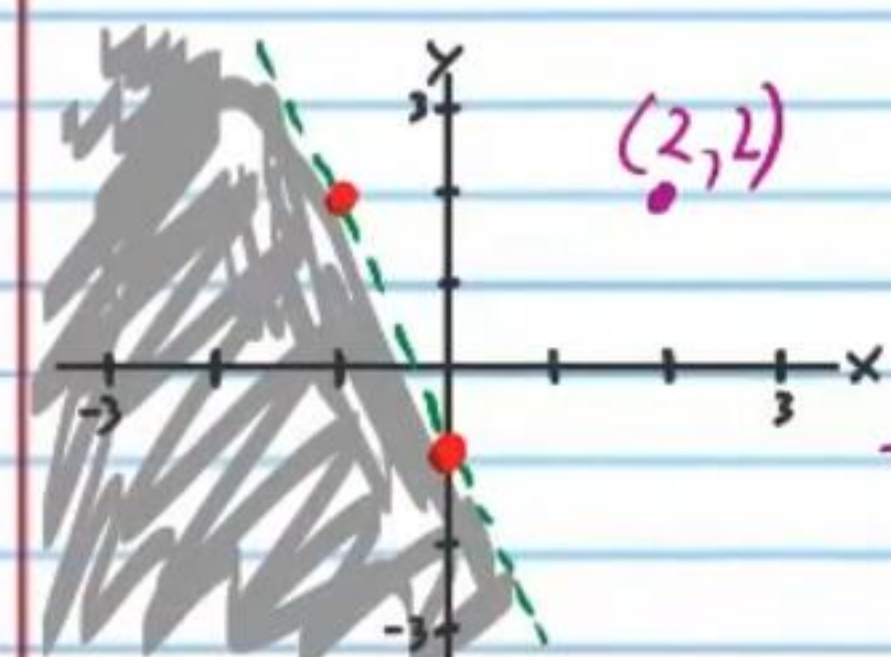
1st Point

$y = 0$
 $\Downarrow$
 $-1 + 2(0) = x$
 $-1 = x$
 $(-1, 0)$

2nd Point

$y = 1$
 $\Downarrow$
 $-1 + 2(1) = x$
 $-1 + 2 = x$
 $1 = x$
 $(1, 1)$

④⑥ $-3x - 1 > y$



x	y
0	-1
-1	2

Test Point
 $-3(2) - 1 > 2$
 $-6 - 1 > 2$
 $-7 > 2$

1st Point

$x = 0$
 $\Downarrow$
 $-3(0) - 1 = y$
 $0 - 1 = y$
 $-1 = y$
 $(0, -1)$

2nd Point

$x = -1$
 $\Downarrow$
 $-3(-1) - 1 = y$
 $3 - 1 = y$
 $2 = y$
 $(-1, 2)$

④⑦ $-2 - \frac{y}{5} \geq 3$

